Collaborative Dual-Branch Network Based on Information Propagation Graph Convolution for Few-Shot Hyperspectral Image Classification

Wenqing Wang, *Member, IEEE*, Pingping Zhang, and Han Liu, *Senior Member, IEEE*

***Abstract*—Graph convolutional networks (GCNs) involve extensive matrix operations, which significantly increase training and inference times and can easily lead to overfitting. In addition, the performance of existing GCN-based methods heavily relies on the quality of node features. When the initial features fail to represent the information of nodes and edges accurately, it becomes difficult to achieve satisfactory classification performance. To address these challenges, we propose a collaborative dual-branch network based on information propagation graph convolution, referred to as information-propagation graph convolutional collaborative dual-branch network (IPGCNet). IPGCNet establishes a task-oriented complementary interaction mechanism between pixel-level refinement and regional-level graph reasoning. The upper branch employs multiscale deep separable convolution and attention mechanism (MSDCA). It achieves pixel-level geometric and spectral compensation by combining spectral band grouping attention and geometrically aware position attention, and can restore the fine detail lost during superpixel aggregation. The lower branch constructs a multiscale graph convolution module employing a four-step optimization strategy, including multiscale fusion adjacency matrix (MFAM), graph propagation mechanism (GPM), channel attention module (CAM), and global supervision module (GSM). This design enables precise modeling of complex internode relationships, thereby improving the accuracy and stability of processing graph-structured data. We evaluated the proposed method on five widely used and representative public hyperspectral image datasets. Experimental results demonstrate that IPGCNet achieves superior classification performance in few-shot settings and significantly outperforms state-of-the-art methods in overall accuracy (OA).**

***Index Terms*—Attention mechanism, convolutional neural network (CNN), graph convolutional network (GCN), hyperspectral image classification.**

I. Introduction

HYPERSPECTRAL imaging (HSI) technology is capable of recording not only spatial information, such as the spatial location, geometric shape, and texture structure of targets, but also spectral radiation data across hundreds of continuous bands. The spectral resolution of HSI can reach the nanometer level, covering a wide range of electromagnetic spectrum bands, including visible light, near-infrared, mid-infrared, and thermal infrared [1], [2]. Each surface material exhibits a unique spectral fingerprint, characterized by distinct reflection, absorption, and emission properties across different spectral bands. By leveraging its rich spectral dimension, HSI can precisely capture subtle spectral differences among diverse surface materials, thereby revealing their physical and chemical properties as well as material composition.

In practical applications, hyperspectral image classification plays a crucial role in various fields, including environmental monitoring [3], geological exploration [4], [5], and agriculture [6], [7], [8], benefiting from these spectral and spatial characteristics. However, hyperspectral image classification remains a challenging task in remote sensing information processing due to the high dimensionality and complexity of HSI data. It is essential to adopt advanced techniques, such as spectral–spatial joint modeling [9], [10] and deep learning (DL) [11], [12], [13], to overcome the limitations of traditional methods and achieve higher accuracy.

Early HSI classification methods primarily relied on traditional machine learning techniques based on statistical learning theory. These methods, including support vector machines (SVMs) [14], random forests (RFs) [15], and K-nearest neighbors (KNNs) [16], depended heavily on handcrafted features and meticulous parameter tuning. While traditional approaches exhibit advantages such as robustness with limited samples and higher interpretability, they generally lack the capacity to learn complex, high-dimensional feature representations, often failing to fully capture the intricate spectral–spatial patterns embedded in HSI data.

To address these challenges, DL techniques have been introduced into HSI classification. DL models, such as convolutional neural networks (CNNs) [17], [18], [19], recurrent neural networks (RNNs) [20], [21], and transformer-based architectures [22], [23], consist of multiple hidden layers and a large number of neurons. These models enable automatic feature extraction and can learn complex nonlinear relationships from high-dimensional data, thereby significantly enhancing classification performance. However, CNNs are inherently constrained by their local receptive fields, which limit their

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Wenqing Wang and Han Liu are with Shaanxi Key Laboratory of Complex System Control and Intelligent Information Processing and the School of Automation and Information Engineering, Xi'an University of Technology, Xi'an 710048, China (e-mail: wangwenqing@xaut.edu.cn; liuhan@xaut.edu.cn).

Pingping Zhang is with the School of Automation and Information Engineering, Xi'an University of Technology, Xi'an 710048, China (e-mail: 2240321313@stu.xaut.edu.cn).

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ability to capture long-range contextual relationships and the non-Euclidean interactions commonly present among hyperspectral pixels. While multiscale CNN variants such as FCN [24], MSPN [25], and lightweight models like 3-D-LWNet [26] enhance local feature capture and reduce overfitting, they still lack explicit modeling of global region-level structures. Furthermore, CNNs rely heavily on patch-based sampling, making them less robust under limited training data and prone to texture distortions or inconsistent predictions across spatial neighborhoods. More advanced global modeling architectures have been proposed to alleviate these limitations. For instance, Xu et al. [27] introduced DSFormer to enhance spectral–spatial interaction through selective fusion strategies. Luo et al. [28] proposed MMD-MLP to capture local–global dependencies via multiresolution directional representations. Xu et al. [29] presented MambaMoE, which incorporates mixture-of-experts mechanisms within state-space models to improve spectral–spatial adaptability while maintaining linear computational complexity. Recently, foundation models such as SpectralGPT [30] and GFM [31] have explored large-scale spectral–spatial pretraining for hyperspectral representation learning. Although these methods can improve the aggregation of global features and sequence modeling, they do not explicitly capture the non-Euclidean relationships between pixels and regions. Therefore, it is still very important to explicitly model the structural relationships between pixels and regions for hyperspectral image classification.

To explicitly model non-Euclidean relational structures, graph convolutional networks (GCNs) [32] have been introduced into hyperspectral image classification. By performing feature propagation on the constructed graph structure, GCN-based methods model long-range dependencies across pixels and regions. In recent years, these methods have evolved along multiple technical routes, including graph structure construction, feature propagation strategies, multiscale context modeling, and attention mechanisms. MiniGCN [33] alleviates the computational burden of full-graph operations through a mini-batch training strategy and achieves efficient processing of large-scale hyperspectral data via local neighborhood sampling. Similarly, Pu et al. [34] construct an adjacency matrix using local spectral–spatial data blocks, enabling the model to perform feature reasoning based on localized consistency. The core advantage of these methods lies in their computational efficiency. However, because they rely solely on local neighborhoods for graph construction, they struggle to capture global relationships across regions and are susceptible to noise within local neighborhoods. Ding et al. [35] propose GCGCN, which constructs a structural topology that better aligns with object boundaries through superpixel segmentation, thereby ensuring global structural consistency and significantly enhancing semantic coherence within object regions. Jia et al. [36] propose graph-in-graph (GiG), which integrates a two-layer structure consisting of an outer graph and an inner graph, and utilizes nested graph convolution to strengthen feature fusion across scales and regions. These methods capture broader contextual information and remain highly sensitive to regional structures, but their performance depends on the quality of superpixels or graph construction; when the adjacency matrix lacks expressive power, misleading feature propagation may occur.

With the development of graph attention mechanisms, numerous studies have attempted to introduce the self-attention structure into GCNs. WFCG [37] enhanced the selectivity of neighborhood aggregation by combining CNN with GAT and adaptively adjusting neighbor weights. Zhou et al. [38] proposed the attention multihop graph and multiscale convolution fusion network (AMGCFN), which modeled deeper contextual semantics through multihop graph attention and multiscale convolution in parallel. Dai et al. [39] developed the multirelational graph attention network (MRGAT), which introduces a multirelation graph structure and adaptively learns the importance weights of different edge types. Xi et al. [40] designed the cross-scale graph prototypical network (X-GPN) to provide robust semantic guidance by relying on cross-scale prototype representations and reducing noise propagation. Xu et al. [41] presented the spectral–spatial residual graph attention network (S2RGANet), which strengthens the response of key bands by using spectral residuals and attention modules. These methods enhance the explicit modeling ability of graph convolution through the attention mechanism, but their computational complexity and dependence on the quality of initial features also increase.

Despite these advancements, GCNs still face numerous challenges [42], [43], [44]. In the graph convolution operation, GCNs require a large number of matrix multiplications and additions, which consume a significant amount of memory and computing resources, thereby increasing the training and inference time and potentially leading to overfitting. In addition, the performance of GCNs largely depends on the quality of the initial node features [45]. If these features fail to accurately represent the information of nodes and edges, the resulting classification or prediction performance may significantly deteriorate [46], [47]. Moreover, most GCN models rely on fixed adjacency matrices, making it difficult to simultaneously capture multidimensional relationships such as structural topology, spatial position, and spectral similarity. Therefore, we propose an information-propagation graph convolutional collaborative dual-branch network (IPGCNet). The entire network consists of two complementary feature extraction branches. For the upper branch, the multiscale deep separable convolution and attention mechanism (MSDCA) module is employed to extract fine-grained spatial-spectral features. This module utilizes multiscale deep convolution, channel grouping attention based on spectral band structure, and position attention with geometric perception to enhance detail expression and spatial consistency. For the lower branch, a graph structure is constructed based on the multiscale fusion adjacency matrix (MFAM), integrating structural topology, spatial distance, and feature similarity. Combining the information propagation mechanism (GPM), the channel attention module (CAM), and global supervision, a four-step optimization strategy is formed to achieve stable multihop information propagation and regional semantic enhancement.

Overall, the main contributions of this article are summarized as follows.

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1) A task-driven pixel-level enhancement branch that complements graph reasoning. Instead of serving as a standalone CNN backbone, the proposed MSDCA module is designed as a functional compensation mechanism to address the intrinsic information loss of superpixel-based graph propagation. MSDCA restores pixel-level fine textures, geometric structures, and spectral details that are inevitably weakened during regional averaging. Employing multiscale depthwise convolution, spectral-band grouping attention, and geometry-aware positional attention, it provides fine-grained features unavailable to the graph branch. This establishes a complementary and interdependent dual-branch architecture, rather than a simple parallel fusion, enabling robust and holistic spectral and spatial representation learning.
2) To address the limitation that traditional GCNs rely on a single adjacency matrix and struggle to adapt to dynamic changes in graph structures, we propose a MFAM construction method. This method integrates structural topology, spatial position, and feature similarity information. By employing different top-k strategies, it extracts graph node features at multiple scales, thus generating adjacency matrices with stronger global representation and fine-grained descriptive capabilities. This approach enables more accurate modeling of complex node relationships and improves both accuracy and stability in processing graph-structured data.
3) To achieve deep feature mining at the superpixel level, we design a feature extraction framework that integrates multidimensional information. Specifically, we introduce a graph information propagation iteration mechanism to optimize information propagation paths and enhance node feature aggregation. Meanwhile, an embedded channel attention enables the model to focus on salient features. Combined with a global supervision module (GSM) to impose structural constraints on features, this forms a four-step optimization strategy together with MFAM. This strategy facilitates efficient fusion of local detailed information and global semantic representations.

The remainder of this article is organized as follows. Section II introduces the proposed IPGCNet; Section III explains the experiments; and the conclusion is presented in Section IV.

II. Proposed Methodology

In this section, we introduce the developed IPGCNet method in detail. The complete framework of the proposed model is illustrated in Fig. 1. In the upper branch, the hyperspectral image first passes through a preprocessing module that uses two convolutional layers to extract preliminary features and perform dimensionality reduction. The resulting features are then fed into a double MSDCA module, which consists of channel attention and position attention mechanisms based on depthwise separable convolutions with different kernel sizes. Simultaneously, the image is also input to a superpixel segmentation module to generate a feature matrix and adjacency matrix for constructing the graph structure. In the lower branch, through a four-step optimization strategy, the node features capture local and global information at multiple scales using the MFAM. Then, the GPM is used to enable information interaction among nodes, followed by the CAM, which further emphasizes important features. Finally, a GSM is applied within the GCN to further optimize the node representations. Ultimately, the features from the pixel-level branch and the superpixel-level branch are fused using a weighted addition strategy. Next, we provide a detailed explanation of each module.

A. Preprocessing Module

HSI contains abundant spectral and spatial information; however, they are also affected by various types of noise, which can lead to classification errors and degraded performance. Therefore, preprocessing is essential to denoise and reduce the dimensionality of the data, thereby enhancing its overall quality and robustness.

This preprocessing module primarily consists of two convolutional layers. Each layer begins with batch normalization to accelerate the training process and improve model stability. Subsequently, a 1×1 convolutional kernel is employed to reduce the number of channels, achieving a linear combination of features while maintaining spatial consistency. Nonlinearity is then introduced through an activation function, enabling the model to learn complex and discriminative feature representations. This sequence of operations is repeated in the second layer to further refine the features.

After the convolutional operations, the output feature dimensions are adjusted to match the height and width of the input image, ensuring seamless integration with subsequent modules. Specifically, by feeding the features $X_{h,w,c}$ into the spectral convolutional layer, the operation is expressed as

$$\hat{X}_{h,w,m} = sig\left(\sum_{m} W_{i,j,m}\, BN\left(X_{h,w,c}\right) + B_{i,j,m}\right) \tag{1}$$

where $X_{h,w,c}$ denotes the input feature map of the HSI, and h, w, c, and m represent the height, width, number of input channels, and number of output channels, respectively. $W_{i,j,m}$ is the weight matrix of the convolutional kernel with a shape of $[m, c, 1, 1]$, and $B_{i,j,m}$ is the corresponding bias vector. $BN(\cdot)$ denotes batch normalization, while σ is the LeakyReLU activation function. The resulting $\hat{X}_{h,w,m}$ is the output feature map, which serves as the denoised and dimensionally reduced representation of the input.

B. Superpixel Segmentation Module

Before extracting superpixel features, the SLIC with linear discriminant analysis (SLIC-LDA) algorithm [48] is used for superpixel segmentation, and the SLIC algorithm is an improved method of the SLIC algorithm. We use LDA to

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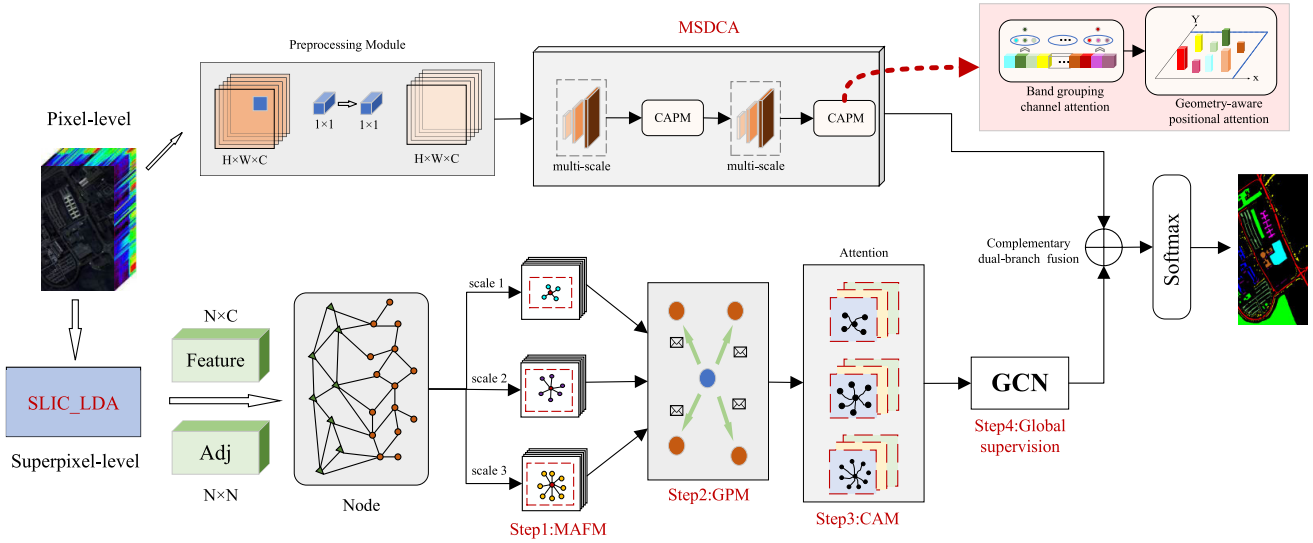


Fig. 1. Architecture of the proposed IPGCNet.

perform discriminant analysis on superpixel features, which can improve the accuracy of superpixel segmentation and classification performance, while having better robustness to data noise and variation. The SLIC-LDA algorithm can be understood as a two-stage process.

Phase One (SLIC Superpixel): First, the segmentation process is initialized by uniformly selecting a set of initial points on the image at fixed intervals. These points are then refined by searching within their local neighborhood for positions with minimal surrounding gradient, ensuring they serve as more representative cluster centers. For each pixel, an integrated distance to every cluster center is calculated, which simultaneously considers both spectral similarity and spatial proximity. Each pixel is assigned to its nearest cluster center, thereby forming initial superpixel blocks.

For each superpixel block, the cluster center is updated iteratively. The spectral vector of the new center is computed as the mean of the spectral vectors of all pixels within that block, while its spatial coordinates are updated as the mean positions of the contained pixels. Specifically, we have

$$H_k^{(t+1)} = \frac{1}{N_k} \sum_{P \in C_k} H_p \tag{2}$$

$$x_k^{(t+1)} = \frac{1}{N_k} \sum_{P \in C_k} x_p \tag{3}$$

$$y_k^{(t+1)} = \frac{1}{N_k} \sum_{P \in C_k} y_p \tag{4}$$

where $H_k^{(t+1)}$ denotes the updated spectral vector of the kth cluster center, $x_k^{(t+1)}$ and $y_k^{(t+1)}$ are the updated spatial coordinates, C_k represents the kth superpixel block, and N_k is the number of pixels in C_k.

Phase Two (LDA Feature Extraction and Discriminant Analysis): The primary goal of LDA is to derive a projection matrix W that maximizes the ratio of interclass scatter to intraclass scatter, thereby enhancing class separability. The intraclass scatter matrix S_W quantifies the compactness of feature vectors within each class and is defined as

$$S_W = \sum_{i=1}^{c} \sum_{j=1}^{n_i} \left(f_j^{(i)} - \mu_i\right) \left(f_j^{(i)} - \mu_i\right)^T \tag{5}$$

where c denotes the number of classes, n_i is the number of superpixel blocks in class i, $f_j^{(i)}$ represents the feature vector of the j_{th} superpixel block in class i, and μ_i is the mean feature vector of class i. The interclass scatter matrix S_B evaluates the separability between class centers and is defined as

$$S_B = \sum_{i=1}^{c} n_i \left(\mu_i - \mu\right) \left(\mu_i - \mu\right)^T \tag{6}$$

where μ denotes the global mean vector of all superpixel blocks. By solving for W, the original high-dimensional superpixel features are projected into a lower dimensional subspace that maximizes interclass variance while minimizing intraclass variance. The resulting projected features retain enhanced discriminative information, providing an optimized representation for subsequent classification or image analysis tasks.

C. MSDCA

The MSDCA module is designed to simultaneously model the multiscale spatial structure and band-related spectral features of hyperspectral images in the pixel domain. It also utilizes the series connection of channel grouping attention and position attention to perform feature selection and enhance the spatial structure. In summary, this module consists of three main components: multiscale depthwise separable convolution, channel grouping attention based on spectral bands, and position attention with geometric perception. The combination of these three groups can effectively enhance the feature expression ability in the pixel domain, maintain a relatively low computational complexity, and provide supplementary stable

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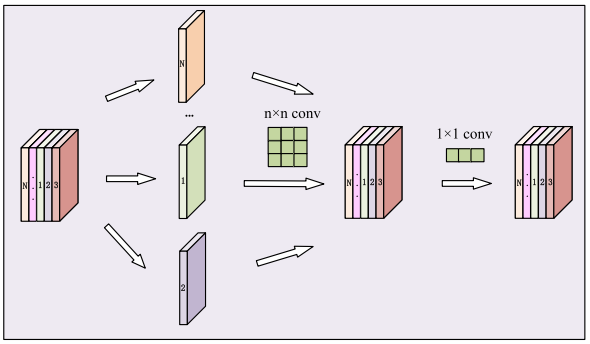


Fig. 2. Schematic of depthwise separable convolution. Each input channel is convolved independently by its corresponding depthwise kernel, followed by a point-wise convolution to integrate interchannel information.

and identifiable features for the subsequent graph information propagation branch.

For the input features $X \in \mathbb{R}^{C\times H\times W}$, the channel transformation is first performed through a 1×1 point convolution. We complete the linear combination of channels without changing the spatial scale, which helps extract spectral mixture features in the continuous bands of hyperspectral data. This approach can reduce the input dimension of the subsequent deep convolutional layers, lower the computational cost, and enable multiscale convolution to be applied to deeper feature representations. Based on this, the module introduces three depthwise separable convolution layers with distinct kernel sizes of 3×3, 5×5, and 7×7, thereby enabling multiscale feature extraction. The structural workflow of depthwise separable convolution is schematically illustrated in Fig. 2. The outputs of the three different scales, $X_{3\times3}$, $X_{5\times5}$, and $X_{7\times7}$, are concatenated along the channel dimension and subsequently refined by a final point-wise convolution to form the aggregated multiscale feature map X, which is formally defined as

$$X_{ms} = \mathrm{Conv}_{1\times1}\left(\mathrm{Concat}\left(X_{3\times3}, X_{5\times5}, X_{7\times7}\right)\right). \tag{7}$$

This design strategy effectively captures rich semantic and detailed information across different spatial scales, furnishing robust feature representations for subsequent processing stages.

To refine the multiscale convolutional features, the MSDCA module incorporates a spectral–geometric attention mechanism that extends the original channel–position attention (CPAM) with hyperspectral-specific priors. Unlike conventional attention that treats all channels and all spatial positions uniformly, the proposed refinement module explicitly models the physical structure of the spectral dimension and the geometric regularity of spatial layouts.

1) Band Grouping Channel Attention: Given the fused feature map $X_{ms} \in \mathbb{R}^{C\times H\times W}$, the spectral channels are not independent as in RGB images, but instead exhibit strong correlations within continuous wavelength intervals. To respect this physical property, the C channels are first divided conceptually into G spectral groups

$$X_{ms} = \left\{X_{ms}^{(1)}, X_{ms}^{(2)}, \ldots, X_{ms}^{(G)}\right\}. \tag{8}$$

Within each spectral group, global average pooling (GAP) and global max pooling (GMP) are applied to extract compact global descriptors

$$\mathbf{z}_{avg}^{(g)} = \mathrm{GAP}\left(X_{ms}^{(g)}\right), \quad \mathbf{z}_{max}^{(g)} = \mathrm{GMP}\left(X_{ms}^{(g)}\right). \tag{9}$$

These descriptors are then passed through the shared MLP of the standard channel-attention formulation

$$a_g = \sigma\left(\mathrm{MLP}\left(\mathbf{z}_{avg}^{(g)}\right) + \mathrm{MLP}\left(\mathbf{z}_{max}^{(g)}\right)\right) \tag{10}$$

where the MLP serves as a learnable band-group reweighting function. The group-wise attention weight is broadcast to its corresponding spectral group

$$X_{cg}^{(g)} = a_g \otimes X_{ms}^{(g)} \tag{11}$$

and all refined groups are concatenated to obtain the channel-enhanced feature

$$X_{cg} = \mathrm{Concat}\left(X_{cg}^{(1)}, \ldots, X_{cg}^{(G)}\right). \tag{12}$$

2) Geometry-Aware Positional Attention: Hyperspectral scenes typically exhibit strong spatial regularity and region-level geometric structure. To incorporate explicit geometric features, normalized spatial coordinate embeddings are constructed as

$$E_{xy}(i,j) = \left[\frac{i}{H}, \frac{j}{W}\right] \tag{13}$$

and concatenated with the channel-refined features

$$X_{geo} = \mathrm{Concat}\left(X_{cg}, E_{xy}\right). \tag{14}$$

Following the structure of the original position-attention module, two consecutive 1×1 convolutions are applied to generate the spatial attention map.

$$M_p = \sigma\left(\mathrm{Conv}_{1\times1}\left(\mathrm{Conv}_{1\times1}\left(X_{geo}\right)\right)\right) \tag{15}$$

and the final spatially refined output is computed as

$$X_{\mathrm{MSDCA}} = M_p \otimes X_{cg}. \tag{16}$$

By embedding spectral-group priors and coordinate-based geometric cues into the established attention structure, the proposed MSDCA module effectively enhances both spectral discriminability and geometry-consistent feature responses.

D. MFAM

In complex graph data structures, node relationships naturally form multidimensional and intricately interwoven networks. Solely relying on structural connectivity significantly limits the semantic and spatial richness embedded within node features, inevitably leading to the omission of critical contextual information. This limitation hinders the model's ability to comprehensively and accurately capture the intrinsic logic and latent patterns present in the data during the learning process.

To overcome this challenge, it is essential to construct a fused adjacency matrix that effectively integrates feature similarity, spatial proximity, and structural connectivity. Unlike a single adjacency matrix, the fused adjacency matrix combines multiple dimensions of node relationships, thereby establishing a multiperspective relational network. Such a construction

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allows node associations to be analyzed and interpreted from both local and global scales, resulting in a more faithful representation of the complex internal relationships inherent in the graph.

1) Feature Similarity Matrix Construction: The node feature matrix X is first normalized to ensure fair comparison. The cosine similarity between nodes i and j is computed as

$$\text{feat}_{ij} = X_i \cdot X_j^T \tag{17}$$

where feat_{ij} denotes the feature similarity score between node i and node j. Subsequently, a Gaussian kernel is applied to emphasize nodes with high similarity and attenuate dissimilar ones, resulting in the feature similarity weighting matrix

$$A_1(i,j) = \exp\left(-\frac{\left|\text{feat}_{ij} - 1\right|}{2{\sigma_g}^2}\right) \tag{18}$$

where σ_g is a hyperparameter controlling the decay rate. Spectral feature similarity is used to capture regions that are spatially nonadjacent but spectrally correlated. For example, the same material often shows fragmented distribution in remote sensing images. By connecting these spectrally similar but spatially discontinuous nodes, the model can better learn the spectral discrimination information of the material, and is particularly suitable for small sample scenarios.

2) Spatial Similarity Matrix Construction: A coordinate matrix coords $\in \mathbb{R}^{N\times 2}$ is introduced, where N represents the number of nodes, and each row encodes the 2-D spatial coordinates of a node. The Euclidean distance between nodes i and j is calculated as

$$\text{dis}_{ij} = \sqrt{\sum_{k=1}^{2}\left(\text{coords}_{ik} - \text{coords}_{jk}\right)^2}. \tag{19}$$

Using this, the spatial similarity matrix is derived as

$$A_2(i,j) = \exp\left(-\frac{\text{dis}_{ij}^2}{2{\sigma_g}^2}\right). \tag{20}$$

Spatial proximity reflects local spatial continuity, which is a fundamental property of remote sensing scenes. By strengthening connections between geographically close nodes, spatial similarity promotes local smoothness, suppresses impulsive noise, and stabilizes label propagation in homogeneous regions.

3) Structural Similarity Matrix Construction: The original adjacency matrix A is directly used to represent structural connectivity, denoted as A_3. The topological structure derived from superpixel segmentation retains the semantic boundaries at the regional level. It ensures that the nodes belonging to the same land cover object remain closely connected and respect the boundaries of different categories. This structural clue prevents the graph from being dominated by pixel-level variations in noise.

4) Top-k Filtering and Neighbor Selection: After obtaining A_1, A_2, and A_3, a top-k operation is applied to each matrix to selectively retain only the most informative neighbors for each node. Specifically, for each node v_i, only the k neighbors with the highest similarity scores are kept, while the remaining entries are set to zero. This selective filtering ensures that the model focuses on the most relevant and meaningful local or global connections, thereby reducing noise and mitigating the risk of overfitting.

Notably, different k values allow flexible multiscale representation: smaller k values highlight fine-grained local structures and subtle patterns, whereas larger k values emphasize broader global relationships and contextual dependencies. Thus, the choice of k controls the tradeoff between local detail preservation and global context awareness.

The topological structure derived from superpixel segmentation retains the semantic boundaries at the regional level. It ensures that the nodes belonging to the same land cover object remain closely connected, while respecting the boundaries of different categories. This structural clue prevents the graph from being dominated by pixel-level variations in noise.

Finally, all similarity matrices are symmetrized and normalized to prevent numerical instability during training. Let

$$\hat{A}_i = D_i^{-1/2} A_i D_i^{-1/2}, \quad i \in \{1,2,3\} \tag{21}$$

denote the symmetrically normalized adjacency matrices, where D_i is the corresponding degree matrix. The fused adjacency matrix is then defined as

$$A_{\text{fuse}} = \hat{A}_1 + \hat{A}_2 + \hat{A}_3 \tag{22}$$

which explicitly incorporates normalization into each component adjacency matrix. This comprehensively captures multidimensional relationships among nodes and supports robust, adaptive feature aggregation.

By integrating these complementary similarities, MFAM constructs an expressive multirelational adjacency matrix that jointly models regional topology, spatial continuity, and spectral discriminability.

E. GPM

The multiscale fusion adjacency matrices constructed by MFAM are designed to capture node connectivity information across different scales, providing a rich structural context. Specifically, for each scale, a fused adjacency matrix A_{fuse} is constructed by integrating feature similarity, structural topology, and spatial proximity. These adjacency matrices are then individually input into the GPM, resulting in multiscale feature representations X_1, X_2, and X_3. The information propagation process of the GPM is shown in Fig. 3.

To integrate the multiscale information, we compute an average of the features from different scales to obtain a comprehensive composite feature X. This averaged feature is then added to the initial residual feature $H^{(0)}$, forming a final enriched representation. Such a design alleviates the vanishing gradient problem commonly encountered in deep graph networks and improves convergence stability during training. Moreover, it supports robust aggregation of both local and global structural information.

Traditional graph neural networks (GNNs) often suffer from insufficient feature propagation, which limits the utilization of global context within the graph. To address this limitation, the proposed GPM introduces a multihop propagation strategy combined with residual connections. Through multiple iterations, node features are propagated across the graph, enabling

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each node to aggregate information not only from immediate neighbors but also from distant nodes, thereby enriching the contextual representation.

The iterative feature propagation process is formulated as follows:

$$H^{(k+1)} = (1-\alpha) A_{\text{fuse}} H^{(k)} + \alpha H^{(0)} \tag{23}$$

where

$A_{\text{fuse}} \in \mathbb{R}^{N\times N}$	fused adjacency matrix at each scale, encoding multidimensional similarity;
$H^{(k)} \in \mathbb{R}^{N\times C}$	node feature matrix after k iterations;
$H^{(0)} \in \mathbb{R}^{N\times C}$	initial node feature matrix, serving as a residual connection;
$\alpha \in [0,1]$	tradeoff parameter that balances the contribution between propagated features and original features.

The parameter K denotes the total number of propagation iterations (propagation depth). When α is set closer to 1, more original information is retained, whereas a smaller α emphasizes aggregated neighbor information.

After K iterations, the final node feature representation is obtained by averaging the intermediate representations

$$X = \frac{1}{K}\sum_{k=1}^{K} H^{(k)}. \tag{24}$$

Finally, to enhance feature discriminability and mitigate oversmoothing, the composite feature X is combined with the initial feature as

$$X_{\text{final}} = X + H^{(0)}. \tag{25}$$

This design effectively allows each node to iteratively collect and integrate information from an expanding receptive field, while maintaining its original characteristics. As a result, the expressive power of node representations is significantly enhanced, leading to improved classification accuracy and robustness. The effectiveness of this multihop residual propagation mechanism is validated through extensive experiments on multiple hyperspectral image datasets.

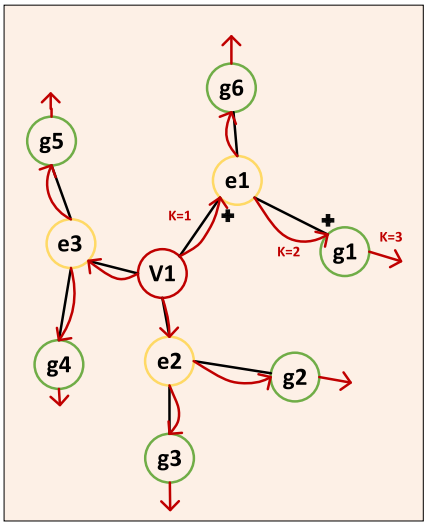


Fig. 3. Schematic of GPM. Circles (v_1, e_1, and g_1) represent nodes, and lines represent edges describing node relationships. Nodes e_1, e_2, and e_3 are directly connected to the target node v_1, while g_1–g_6 denote more distant nodes reached through multihop propagation.

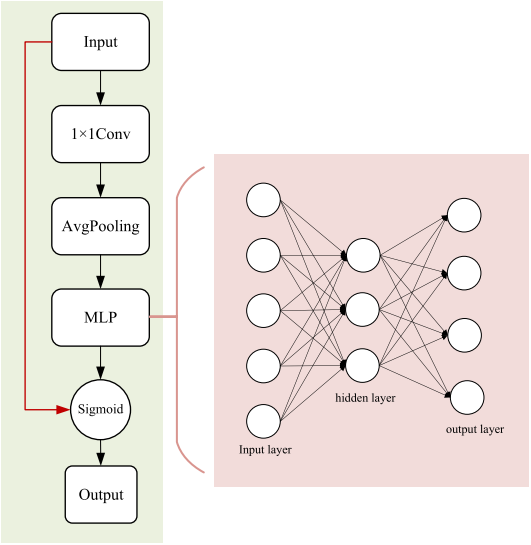


Fig. 4. Architecture of the CAM. It emphasizes important channels and suppresses irrelevant information through adaptive weighting.

F. CAM and GSM

After multihop feature propagation via the GPM, although richer contextual information is integrated, some noisy or less informative features may also be inadvertently introduced. Such noise can dilute the discriminative power of critical features and adversely affect model performance. To mitigate this issue, we introduce the CAM, which selectively emphasizes informative feature channels while suppressing irrelevant or noisy ones. By adaptively assigning different importance weights to each channel, CAM guides the model to focus on the most relevant feature dimensions, thereby enhancing the robustness and discriminative ability of node representations. The structural diagram of CAM is depicted in Fig. 4, illustrating how it refines feature importance at the channel level.

Specifically, let the feature matrix after GPM be denoted as $H \in \mathbb{R}^{N\times C}$, where N represents the number of nodes and C denotes the number of feature channels. The channel attention mechanism first computes a global channel descriptor $\mathbf{z}$ by averaging over all node features

$$\mathbf{z} = \frac{1}{N}\sum_{i=1}^{N} H_i \tag{26}$$

where $H_i \in \mathbb{R}^C$ denotes the feature vector of node i. This descriptor $\mathbf{z}$ captures the overall feature importance across all nodes.

Subsequently, $\mathbf{z}$ is passed through a multilayer perceptron composed of two fully connected layers with a ReLU activation in between. After processing by the MLP, a sigmoid activation function is applied to produce the final channel-wise attention weights. We have

$$\mathbf{a} = sig\,(\text{MLP}\,(\mathbf{z})) \tag{27}$$

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Fig. 5. Detailed information of the five datasets. (a1)–(a5) Ground reality maps of the PaviaU, Salinas, WHU-Hi-HongHu, WHU-Hi-LongKou, and WHU-Hi-HanChuan datasets. (b1)–(b5) Real color maps.

where $\sigma(\cdot)$ denotes the sigmoid function, ensuring that the attention weights are constrained within the range $[0, 1]$.

The refined output feature matrix H_{att} is then obtained via element-wise multiplication between the original feature matrix H and the attention weights $\mathbf{a}$, broadcast across all nodes

$$H_{\text{att}} = H \odot \mathbf{a} \tag{28}$$

where $\odot$ represents element-wise multiplication. This operation effectively highlights informative channels while suppressing less useful ones, resulting in more robust and expressive node features.

To further enhance global structural awareness and stabilize multihop propagation, we introduce a GSM that imposes an explicit structural constraint on the learned node features. The GSM operates on the attention-refined features H_{att} and applies an additional graph convolution using the symmetrically normalized adjacency matrix A

$$H' = \tilde{A}\ H_{\text{att}} \tag{29}$$

where $\tilde{A}$ is the fused similarity matrix.

Unlike the main GPM, which focuses on local feature refinement, the GSM introduces a global structural prior by propagating H_{att} through the entire adjacency matrix. This encourages nodes belonging to the same semantic region to produce consistent features, while preserving meaningful boundaries encoded in A. Thus, the GSM acts as a structural regularizer that prevents feature drift, suppresses noise, and stabilizes representation learning under few-shot conditions. The output H' is fed into an auxiliary classification head and supervised with the ground-truth labels

$$\mathcal{L}_{\text{gsm}} = \text{CE}\left(\text{Classifier}\left(H'\right),\ Y\right) \tag{30}$$

where CE$(\cdot)$ denotes the standard cross-entropy loss used to measure the discrepancy between the predicted class probabilities and the ground-truth labels. The auxiliary loss applied to H' ensures that the globally aggregated features remain discriminative and structurally consistent. This loss is jointly optimized with the main objective, providing a global structural supervision signal that stabilizes multihop propagation and enhances feature separability.

Together with the CAM, the GSM forms a complementary dual-refinement mechanism. The CAM enhances node features at the channel level, whereas GSM enforces global consistency across the graph. This combination enables the model to capture both local discriminative cues and global relational patterns, resulting in more robust and accurate graph-based classification.

III. Experimental Results

In this section, five HSI datasets were adopted to verify the method we proposed: they are University of Pavia (PaviaU), Salinas, WHU-Hi-HongHu, WHU-Hi-LongKou, and WHU-Hi-HanChuan. The real ground object maps and color maps of the five datasets are shown in Fig. 5.

A. Datasets

1) PaviaU: The PaviaU dataset is provided by the PaviaU in Italy, and the data are collected by the ROSIS [49] sensor. It contains 115 spectral bands, with a spatial resolution of 1.3 m. It covers the campus of the PaviaU in Italy and its surrounding areas, including various types of features such as buildings, roads, and vegetation. Typically, a 610×340 pixel area is used, including nine ground feature categories, with a total sample size of 42 776, and the commonly used marking sample size is 428 categories.

2) Salinas: Provided by Salinas Valley in California, USA, the data was collected by AVIRIS [50] sensors. It contains 224 spectral bands, with a spatial resolution of 3.7 m. It covers an agricultural area, including various types of crops and other ground features. Usually, the data is cropped into an area of 512×217 pixels, containing 16 ground feature categories, with a total sample size of 57 050; however, the commonly marked sample size is only 570 categories.

3) WHU-Hi Series: WHU-Hi-HongHu was collected in Honghu City using a DJI Matrice 600 Pro UAV equipped with a 17-mm Headwall Nano-Hyperspec sensor under cloudy weather conditions with a relative humidity of about 55%. It covers a complex agricultural area with multiple Brassica species and Chinese cabbage varieties. The image size is 940×475 pixels with 270 bands ranging from 400 to 1000 nm, and the spatial resolution is about 0.043 m.

WHU-Hi-LongKou was collected in Longkou Town using a DJI Matrice 600 Pro UAV equipped with an 8-mm Headwall Nano-Hyperspec sensor under clear weather with a relative humidity of about 65%. It represents a relatively simple agricultural scene containing six crop categories, including

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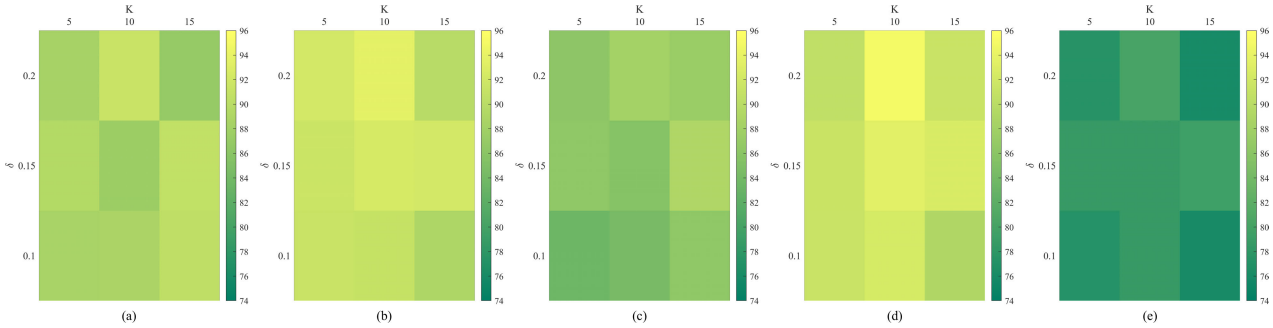


Fig. 6. Relationship between the hyperparameters k, α and the evaluation metric OA. (a) PaviaU. (b) Salinas. (c) WHU-Hi-HongHu. (d) WHU-Hi-LongKou. (e) WHU-Hi-HanChuan.

corn, cotton, sesame, broad-leaved soybean, narrow-leaved soybean, and rice. The image size is 550 × 400 pixels with 270 bands from 400 to 1000 nm, and the spatial resolution is approximately 0.463 m.

WHU-Hi-HanChuan was collected in Hanchuan City using a Leica Aibot X6 UAV equipped with a Headwall Nano-Hyperspec sensor. It corresponds to a rural and urban fringe scene containing buildings, water bodies, cultivated land, and seven crop categories. The image size is 1217 × 303 pixels with 274 bands ranging from 400 to 1000 nm, and the spatial resolution is about 0.109 m.

We adopt three commonly used evaluation metrics, including, overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. Furthermore, we visualize the HSI classification maps obtained from all the compared models for qualitative analysis. Several representative methods are selected for comparative experiments, including CNN-enhanced GCN with pixel-level and superpixel-level feature fusion (CEGCN) [51], GCN and transformer fusion network with spatial–spectral features (GTFN) [52], channel-layer-oriented lightweight network (CLOLN) [53], transformer and graph convolution combining framework (S^2GFormer) [54], groupwise separable convolutional vision transformer network (GSC-VIT) [55], spatial–spectral mamba (MambaHSI) [56], non-Euclidean spectral-spectral feature mining network with gated GCN-CNN (NESSGGCN) [57], and Spatial-Spectral Enhancement and Fusion Network (SSEFN) [58].

For fair comparison, all experiments were conducted using the hyperparameters recommended in the original papers. To eliminate bias introduced by the random selection of training samples, each experiment was repeated ten times. All evaluations were performed under a strict few-shot protocol. In each run, only five labeled samples per class were randomly selected as the training set, while the remaining samples were used for testing. The number of training epochs was set to 200, and the learning rate was fixed at 0.001 for all datasets. All methods were implemented in PyTorch 2.5.1 using Python 3.12, and training was performed on a workstation equipped with an NVIDIA RTX 4090 GPU.

B. Hyperparameter Analysis

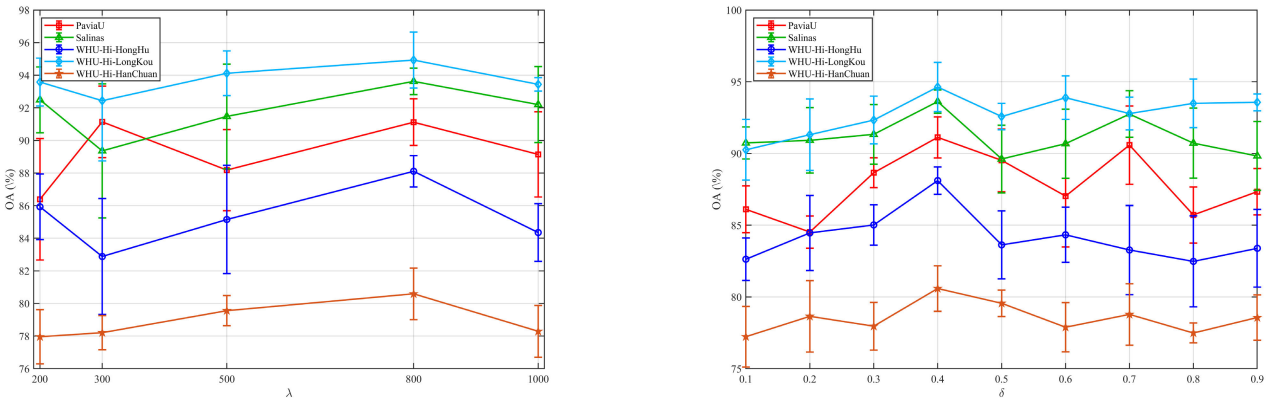
In this section, we analyze the effects of the propagation mechanism parameters k and α, the superpixel segmentation scale λ, and the weighted fusion parameter σ.

1) Analysis of k and α: In order to accurately determine the optimal classification performance of the parameters k and α in the GPM, this article systematically explores their parameter space. Specifically, we selected (k, α) pairs from the set (5, 0.1), (10, 0.1), (15, 0.1), (5, 0.2), (10, 0.2), (15, 0.2). This selection range was carefully designed to cover a reasonable span of values for these parameters, allowing us to comprehensively analyze their effects on classification performance. The relationship between OA and the parameters k (which controls the number of neighboring nodes) and α is illustrated in Fig. 6. By analyzing the results across the five datasets presented in Fig. 6, it is evident that the OA values remain consistently high when α is set to 0.20. This suggests that setting α to 0.20 enhances the efficiency of information propagation and feature extraction in GPM, thereby improving the classification accuracy. Furthermore, for the parameter k, which represents the number of neighboring nodes, the OA of the two datasets reaches its peak when k is set to 10. This indicates that, for these particular datasets, setting $k = 10$ achieves the optimal balance in defining the neighborhood structure for information propagation.

2) Analysis of λ: To systematically determine the optimal superpixel segmentation scale for achieving the best classification performance, we conducted a comprehensive analysis by varying the parameter λ over {200, 300, 500, 800, 1000}. As a key factor governing segmentation granularity, λ directly affects the balance between spectral fidelity and spatial smoothness in superpixel construction. The OA and λ relationship across multiple datasets is presented in Fig. 7. From these results, $\lambda = 800$ consistently emerges as a robust and stable choice for several datasets.

3) Analysis of σ: To optimize the fusion performance between the upper and lower branches of the proposed model, the parameter σ was varied from 0.1 to 0.9. The corresponding OA–σ curve is shown in Fig. 7. Experimental results indicate that as σ increases from 0.1, OA initially rises, suggesting that a moderate increase in the upper branch contribution enhances the complementary interaction between the two branches. Specifically, assigning slightly higher weight to the graph branch strengthens region-level structural modeling and multihop information aggregation, which is particularly beneficial under few-shot settings where pixel-level supervision is limited. However, when σ exceeds 0.4, OA begins to decline. This may be attributed to the excessive dominance of the graph branch, which can introduce spatial smoothing effects

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(a) (b)

Fig. 7. Evaluation of OA as a function of (a) segmentation scale λ and (b) weighted fusion parameter σ.

TABLE I
ABLATION ANALYSIS OF MFAM IN FIVE DATASETS

A_1	A_2	A_3	PaviaU	Salinas	HongHu	LongKou	HanChuan
✓			88.32 ± 3.10	90.51 ± 2.44	86.05 ± 3.08	88.18 ± 2.12	76.23 ± 1.22
	✓		88.48 ± 2.12	91.11 ± 3.29	85.21 ± 2.41	88.66 ± 0.51	75.09 ± 1.98
		✓	88.75 ± 2.31	88.55 ± 2.18	85.63 ± 2.52	89.11 ± 2.25	76.85 ± 0.58
✓	✓		89.61 ± 1.38	90.82 ± 1.42	88.01 ± 1.05	89.36 ± 1.23	77.65 ± 0.74
✓		✓	90.58 ± 1.66	90.13 ± 1.51	87.29 ± 1.46	89.54 ± 1.04	78.15 ± 1.08
	✓	✓	90.67 ± 2.65	92.01 ± 0.68	86.32 ± 0.74	92.79 ± 0.89	78.64 ± 1.67
✓	✓	✓	91.12 ± 1.43	93.62 ± 0.85	88.11 ± 0.96	94.93 ± 1.72	80.59 ± 1.58

and weaken fine-grained spectral–spatial details preserved by the pixel branch. Overall, $\sigma = 0.4$ provides both high accuracy and stable performance across all five datasets, indicating that a balanced collaboration between local discriminability and global structural consistency yields the most effective representation.

C. Ablation Analysis

1) Effectiveness of MFAM: In order to fully verify the effectiveness of the proposed multiscale fusion adjacency matrix method, a detailed comparison of OA is conducted using different adjacency matrices, as shown in Table I. The MFAM framework contains three different adjacency matrices, namely A_1, A_2, and A_3. The performance of using a single matrix or two matrices shows certain variability. Through comprehensive comparison, in most cases, the fused adjacency matrix of MFAM is the most effective on these five datasets. It not only demonstrates the complementarity of the three adjacency matrices but also highlights the advantages of the comprehensive method in improving the overall classification accuracy.

2) Effectiveness of GPM, CAM, and GSM: To verify the effectiveness of the proposed GPM, CAM, and GSM, we design three ablated variants of IPGCNet by sequentially removing GPM, CAM, and GSM from the upper branch of the network. The results of these ablation studies on five datasets are summarized in Table II. As can be observed, the complete model incorporating all three modules consistently achieves higher OA compared to its ablated counterparts

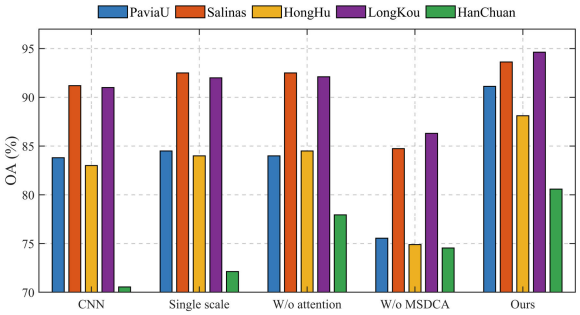


Fig. 8. Ablation analysis of MSDCA in five datasets.

across all datasets. For the w/o GPM setting, the proposed graph propagation mechanism is replaced by a basic single-layer GCN while retaining the remaining graph structure. Notably, removing GPM leads to the most significant performance degradation, indicating that the multihop information propagation mechanism plays a critical role in enhancing structural feature aggregation and semantic consistency. When CAM is removed, the performance drop suggests that channel-wise attention contributes to discriminative feature refinement. Similarly, the exclusion of GSM results in reduced accuracy, demonstrating the importance of global supervision in stabilizing representation learning. These findings strongly support the integration of GPM, CAM, and GSM, underscoring their importance in improving classification accuracy on diverse hyperspectral image datasets.

3) Effectiveness of MSDCA: To thoroughly evaluate the effectiveness of the proposed MSDCA module, we further construct several ablated variants, including a single-scale version that retains only a 3 × 3 kernel, a plain CNN version in which all multiscale operations are removed, a version without the attention mechanism, and a version that completely discards the entire MSDCA module and relies solely on the graph branch for representation learning. The quantitative results on five datasets are presented in Fig. 8. Across all datasets, the single-scale and plain CNN variants exhibit clearly inferior performance, particularly on PaviaU,

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TABLE II
ABLATION ANALYSIS OF GPM, CAM, AND GSM ON FIVE DATASETS

CAM	GPM	GSM	PaviaU	Salinas	HongHu	LongKou	HanChuan
✓			85.71 ± 0.48	90.33 ± 2.89	85.79 ± 1.93	87.31 ± 1.72	75.63 ± 1.84
	✓		86.02 ± 1.09	90.64 ± 4.22	85.10 ± 1.32	87.86 ± 1.04	76.76 ± 1.09
		✓	85.58 ± 1.02	89.11 ± 1.27	84.94 ± 1.44	88.03 ± 1.21	76.05 ± 1.89
✓	✓		89.12 ± 1.38	91.54 ± 1.66	86.73 ± 0.89	90.71 ± 1.11	77.29 ± 1.04
✓		✓	87.36 ± 0.72	91.21 ± 1.62	86.77 ± 0.93	90.59 ± 2.36	78.33 ± 1.11
	✓	✓	87.09 ± 0.94	92.64 ± 1.04	87.61 ± 1.58	89.28 ± 1.08	78.47 ± 1.08
✓	✓	✓	91.12 ± 1.43	93.62 ± 0.85	88.11 ± 0.96	94.93 ± 1.72	80.59 ± 1.58

TABLE III
COMPARISON OF RUNNING EFFICIENCY FOR DIFFERENT METHODS

Method	Running Time(s)								FLOPs(G)
	PaviaU		Salinas		HongHu		LongKou		
SSEFN	670.37	66.05	510.01	854.23	623.55	254.53	698.79	1585.54	15
NESSGGCN	**29.65**	**0.011**	**26.71**	**0.01**	**47.66**	**0.11**	**31.29**	**0.031**	**0.9**
S^2GFormer	80.36	10.37	111.32	17.85	200.50	32.01	100.89	19.52	1.52
our	46.98	0.06	33.85	0.03	66.97	0.17	56.14	0.08	0.92

HongHu, and HanChuan, where the lack of multiscale receptive fields restricts their ability to capture spatial–spectral structures at different granularities. Removing the attention mechanism leads to an even more pronounced degradation, particularly on Salinas and LongKou. In these cases, the OA decreases by roughly 3%–7% relative to the full MSDCA. These results confirm that spectral-band grouping attention and geometry-aware positional attention are both crucial for improving interband discrimination while preserving spatial consistency.

The variant that completely removes MSDCA shows the largest performance decline among all ablations, with accuracy reductions exceeding those of the other variants on every dataset. This demonstrates that pixel-level feature refinement provided by MSDCA is indispensable and cannot be compensated for by the graph branch alone. In contrast, the full MSDCA consistently achieves the highest accuracy across all datasets, outperforming the strongest ablated variant by margins ranging from 2% to over 7%. Overall, these results confirm that both the multiscale depthwise separable convolution and the dual-attention mechanism, as well as the full MSDCA module itself, are indispensable for extracting richer and more discriminative pixel-level features that effectively complement the graph-based branch in the proposed framework.

D. Comparison of Computational Efficiency

In this section, we evaluate and compare the efficiency and computational burden of various methods by analyzing their training and testing times. As shown in Table III, SSEFN generally exhibits higher time consumption across all datasets due to its redundant feature extraction process and complex model architecture, which also demand substantial hardware resources. In contrast, NESSGGCN leverages graph convolution to efficiently capture structural information, thereby reducing computational costs. However, its storage consumption varies across different datasets, and the need to balance global information capture introduces additional computational overhead, resulting in a tradeoff between model complexity and overall efficiency. Benefiting from the lightweight design that integrates multiscale graph convolution and CNN fusion, our proposed method optimizes the feature processing pipeline, reduces redundant computations, and achieves high accuracy while significantly improving operational efficiency. Consequently, our approach demonstrates superior time performance across all evaluated datasets.

E. Classification Results

In this part, we randomly select five labeled samples from each class as the training set. We compare the performance of our proposed method with several representative methods, including CEGCN [51], GTFN [52], CLOLN [53], S^2GFormer [54], GSC-VIT [55], MambaHSI [56], NESSGGCN [57], and SSEFN [58].

A comprehensive examination of the results reported in Tables IV–VIII reveals that the proposed method consistently delivers outstanding performance across five benchmark datasets with diverse characteristics. This demonstrates the strong generalization ability of our model under varying scene structures, class distributions, spectral complexity, and spatial texture variations.

On the PaviaU and Salinas datasets, which represent typical urban and agricultural scenes, the proposed method achieves OA values of 91.12% and 93.62%, respectively. These values are significantly higher than those produced by competing models such as SSEFN, NESSGGCN, and CLOLN. Both datasets exhibit clear spectral differences across land-cover types while maintaining moderate spatial complexity. Our method not only delivers substantial improvements in overall classification accuracy but also achieves nearly perfect class-wise performance. For example, the accuracy for Class 6 on PaviaU reaches 97.32%, and the accuracy for Class 2 on Salinas reaches 100.00%. It is worth noting that although some other methods achieve comparable OA values, their classification maps in Figs. 9 and 10 show boundary fragmentation and isolated classification errors. In contrast, our method generates clearer object boundaries and more spatially consistent regions. This suggests that the proposed dual-branch framework improves not only overall pixel-level accuracy but also spatial consistency and intraclass uniformity, which cannot be fully reflected by OA alone.

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TABLE IV

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR PAVIAU (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	CEGCN	GTFN	CLOLN	S^{2}GFormer	GSC-VIT	MambaHSI	NESSGGCN	SSEFN	Our
1	81.40 ± 9.83	60.06 ± 19.47	88.80 ± 0.66	58.96 ± 5.91	**100.00 ± 0.00**	80.11 ± 10.2	75.66 ± 8.04	55.96 ± 19.61	91.74 ± 4.76
2	82.92 ± 11.76	58.55 ± 7.78	89.11 ± 2.40	74.56 ± 5.94	65.28 ± 4.7	79.59 ± 8.81	78.37 ± 16.02	71.14 ± 13.57	**90.73 ± 2.39**
3	88.25 ± 9.41	73.59 ± 9.48	**88.60 ± 1.06**	84.32 ± 2.43	85.47 ± 1.07	63.89 ± 15.64	80.82 ± 10.99	73.55 ± 9.25	84.85 ± 8.04
4	84.62 ± 4.18	74.14 ± 12.00	78.82 ± 5.71	88.97 ± 3.65	**96.38 ± 1.82**	69.27 ± 8.77	92.18 ± 4.38	80.50 ± 7.72	88.63 ± 6.54
5	**99.95 ± 2.45**	98.73 ± 1.27	99.54 ± 0.45	97.59 ± 1.12	99.63 ± 0.35	99.74 ± 0.69	99.93 ± 0.20	97.24 ± 5.52	99.81 ± 0.23
6	94.89 ± 5.14	85.50 ± 9.06	89.76 ± 3.32	86.12 ± 2.73	96.09 ± 1.90	90.58 ± 5.88	90.00 ± 6.41	73.93 ± 6.90	**97.32 ± 1.71**
7	98.59 ± 2.45	93.09 ± 6.23	91.21 ± 2.43	97.35 ± 2.03	95.52 ± 5.64	81.67 ± 11.01	**99.80 ± 0.20**	71.91 ± 20.37	95.23 ± 7.38
8	90.32 ± 13.38	60.03 ± 9.13	88.34 ± 0.78	44.46 ± 5.97	75.73 ± 3.36	80.52 ± 11.18	85.76 ± 5.97	69.30 ± 20.19	**95.31 ± 5.02**
9	90.50 ± 10.28	92.71 ± 6.51	78.80 ± 0.82	98.64 ± 1.33	81.15 ± 0.48	96.07 ± 4.15	**99.46 ± 0.99**	70.96 ± 9.80	89.87 ± 6.92
OA(%)	86.30 ± 5.66	67.03 ± 8.33	85.29 ± 2.66	74.85 ± 3.56	85.16 ± 2.19	80.40 ± 4.38	82.86 ± 6.58	70.58 ± 5.53	**91.12 ± 1.43**
AA(%)	90.16 ± 3.04	77.38 ± 3.39	88.11 ± 2.24	81.22 ± 2.34	88.36 ± 1.12	82.38 ± 2.44	89.11 ± 1.87	73.83 ± 2.19	**92.61 ± 0.84**
$k \times 100$	82.58 ± 6.84	59.89 ± 8.69	81.06 ± 2.82	68.67 ± 3.61	76.55 ± 2.51	75.94 ± 9.76	78.40 ± 7.48	63.35 ± 6.08	**88.20 ± 1.66**

TABLE V

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR SALINAS (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	CEGCN	GTFN	CLOLN	S^{2}GFormer	GSC-VIT	MambaHSI	NESSGGCN	SSEFN	Our
1	99.78 ± 0.35	99.27 ± 1.06	91.50 ± 5.02	97.67 ± 1.67	**99.99 ± 0.01**	96.36 ± 3.80	98.98 ± 1.21	99.27 ± 1.21	98.50 ± 0.95
2	99.99 ± 0.01	99.06 ± 1.41	84.34 ± 15.08	99.98 ± 0.06	99.65 ± .29	98.22 ± 4.00	**100.00 ± 0.00**	99.67 ± 0.27	**100.00 ± 0.00**
3	**99.91 ± 0.19**	96.38 ± 8.88	98.12 ± 2.99	88.81 ± 7.08	96.96 ± 1.72	94.32 ± 6.22	96.00 ± 10.39	74.78 ± 0.27	95.77 ± 7.30
4	99.72 ± 0.34	98.09 ± 1.17	75.11 ± 13.85	98.06 ± 0.85	99.17 ± 0.96	98.51 ± 2.23	99.36 ± 0.41	96.76 ± 4.03	**99.91 ± 0.10**
5	93.51 ± 4.40	93.98 ± 2.12	62.79 ± 8.35	91.73 ± 1.77	92.27 ± 2.07	96.65 ± 2.09	**97.93 ± 1.23**	95.15 ± 2.82	96.01 ± 2.94
6	99.95 ± 0.36	99.01 ± 1.15	82.20 ± 17.77	98.95 ± 0.87	**100.00 ± 0.00**	99.17 ± 0.77	99.87 ± 0.11	96.42 ± 4.36	97.75 ± 1.82
7	99.89 ± 0.10	97.39 ± 5.96	94.76 ± 9.79	99.08 ± 0.62	98.94 ± 0.10	97.82 ± 2.07	**99.91 ± 0.04**	99.08 ± 1.74	98.72 ± 1.33
8	68.17 ± 17.34	77.45 ± 7.57	65.27 ± 28.84	84.35 ± 1.52	83.87 ± 2.36	77.46 ± 12.01	78.34 ± 11.01	70.49 ± 19.15	**89.01 ± 1.89**
9	**100.00 ± 0.00**	99.18 ± 0.92	81.50 ± 13.66	99.84 ± 0.14	97.53 ± 1.46	89.83 ± 1.51	99.99 ± 0.01	98.68 ± 2.24	99.91 ± 0.08
10	95.23 ± 3.04	92.07 ± 3.98	75.33 ± 19.12	84.41 ± 3.98	74.64 ± 5.84	89.09 ± 9.82	89.88 ± 6.99	84.02 ± 12.02	**95.53 ± 2.05**
11	91.80 ± 10.33	99.44 ± 0.45	66.78 ± 6.83	99.17 ± 0.80	95.06 ± 2.09	98.42 ± 1.95	99.41 ± 0.61	93.72 ± 7.21	**99.81 ± 0.31**
12	**99.92 ± 0.17**	92.77 ± 5.91	82.49 ± 19.85	95.64 ± 1.96	98.96 ± 0.09	99.25 ± 1.07	94.20 ± 1.96	92.14 ± 14.56	98.15 ± 1.94
13	99.15 ± 1.25	92.96 ± 5.40	89.84 ± 10.76	95.07 ± 5.22	**99.88 ± 0.19**	98.71 ± 1.67	99.03 ± 2.07	98.44 ± 1.98	99.08 ± 2.01
14	94.75 ± 4.67	92.85 ± 16.48	84.20 ± 17.74	94.27 ± 1.63	95.65 ± 2.18	**99.13 ± 1.13**	97.24 ± 2.66	96.83 ± 2.82	98.45 ± 1.09
15	88.87 ± 6.32	82.15 ± 6.78	78.28 ± 2.32	72.37 ± 3.84	**93.71 ± 2.10**	89.84 ± 11.68	77.42 ± 10.82	89.25 ± 6.88	80.60 ± 7.31
16	95.52 ± 0.99	**97.20 ± 2.56**	82.73 ± 2.93	93.36 ± 3.87	95.15 ± 0.27	96.02 ± 2.35	92.34 ± 3.67	94.38 ± 5.21	94.12 ± 5.02
OA(%)	92.35 ± 1.89	90.89 ± 1.44	76.44 ± 2.52	90.38 ± 0.57	92.41 ± 0.79	89.08 ± 1.61	89.88 ± 1.72	89.39 ± 2.75	**93.62 ± 0.85**
AA(%)	95.39 ± 1.11	94.33 ± 1.18	80.95 ± 2.14	93.30 ± 0.97	95.09 ± 0.48	94.93 ± 0.55	94.99 ± 1.12	92.44 ± 1.38	**96.33 ± 0.62**
$k \times 100$	91.62 ± 2.09	89.88 ± 1.59	78.96 ± 2.54	89.29 ± 0.64	91.55 ± 0.88	89.13 ± 3.23	88.73 ± 1.90	88.25 ± 1.38	**92.70 ± 0.71**

TABLE VI

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR WHU-HI-HONGHU (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	CEGCN	GTFN	CLOLN	S^{2}GFormer	GSC-VIT	MambaHSI	NESSGGCN	SSEFN	Our
1	78.92 ± 11.42	55.51 ± 19.56	95.05 ± 9.25	91.38 ± 2.05	**98.56 ± 1.99**	80.46 ± 10.40	80.00 ± 9.01	85.31 ± 8.40	80.49 ± 10.82
2	83.69 ± 17.31	74.59 ± 5.63	84.43 ± 4.08	81.60 ± 4.51	75.25 ± 1.97	66.89 ± 11.57	61.73 ± 17.80	**87.34 ± 9.21**	82.75 ± 12.25
3	84.70 ± 6.46	76.82 ± 5.12	83.02 ± 4.41	82.93 ± 1.73	77.76 ± 5.55	82.35 ± 4.81	75.32 ± 11.26	81.17 ± 2.63	**89.14 ± 3.87**
4	88.24 ± 3.91	70.00 ± 10.89	**99.51 ± 0.15**	88.62 ± 3.07	88.46 ± 2.47	84.23 ± 8.66	88.30 ± 12.70	88.30 ± 4.97	94.94 ± 2.08
5	93.56 ± 3.26	77.75 ± 2.96	32.47 ± 8.24	91.47 ± 1.70	85.10 ± 0.42	85.20 ± 8.27	88.43 ± 10.59	93.00 ± 3.35	**95.07 ± 1.58**
6	90.26 ± 4.91	57.71 ± 15.73	**97.61 ± 0.15**	90.93 ± 1.23	89.02 ± 0.28	81.94 ± 7.30	79.64 ± 21.31	92.56 ± 1.22	91.87 ± 0.58
7	61.07 ± 5.15	52.64 ± 12.64	**85.69 ± 4.76**	68.74 ± 3.53	49.38 ± 0.06	53.59 ± 11.64	41.33 ± 18.46	58.10 ± 8.02	68.61 ± 10.47
8	**70.00 ± 16.62**	40.18 ± 20.66	47.97 ± 12.96	43.91 ± 3.95	38.74 ± 1.27	67.31 ± 10.67	38.58 ± 7.07	66.09 ± 8.14	66.87 ± 8.95
9	93.07 ± 0.63	93.38 ± 0.89	93.59 ± 15.44	94.78 ± 1.66	93.06 ± 0.47	86.83 ± 6.44	92.85 ± 6.12	**94.92 ± 2.25**	88.42 ± 9.21
10	62.45 ± 22.74	77.83 ± 9.81	**94.02 ± 0.82**	67.97 ± 3.23	74.78 ± 2.91	57.10 ± 20.34	53.21 ± 15.63	73.08 ± 14.20	85.41 ± 10.28
11	49.17 ± 12.27	59.70 ± 17.22	**79.48 ± 8.13**	50.69 ± 9.63	52.12 ± 9.01	57.50 ± 7.12	38.36 ± 8.80	50.96 ± 13.03	70.15 ± 2.77
12	72.75 ± 5.11	**83.65 ± 4.26**	55.42 ± 7.29	60.38 ± 4.85	68.71 ± 3.97	56.02 ± 18.82	58.00 ± 17.41	67.77 ± 14.60	70.08 ± 5.78
13	65.85 ± 7.43	83.66 ± 6.71	83.54 ± 0.71	**93.62 ± 3.83**	27.49 ± 0.86	68.55 ± 4.00	41.51 ± 9.52	68.30 ± 5.20	80.44 ± 5.01
14	68.96 ± 3.04	**86.73 ± 3.95**	78.54 ± 1.80	67.53 ± 1.32	62.25 ± 6.22	74.04 ± 7.77	63.49 ± 15.58	80.22 ± 6.97	80.26 ± 9.24
15	96.10 ± 1.87	86.09 ± 2.81	52.68 ± 17.80	**98.83 ± 0.44**	93.93 ± 0.05	95.83 ± 2.67	82.84 ± 12.12	97.55 ± 1.61	92.18 ± 3.85
16	77.95 ± 14.44	86.73 ± 2.14	**97.45 ± 1.80**	90.49 ± 1.88	90.93 ± 3.97	76.69 ± 14.60	66.83 ± 31.08	84.42 ± 11.30	83.73 ± 10.95
17	86.32 ± 9.25	72.88 ± 6.25	63.23 ± 5.70	73.60 ± 0.80	58.09 ± 13.59	69.87 ± 16.32	80.22 ± 16.44	81.11 ± 12.36	**89.09 ± 10.18**
18	96.93 ± 2.01	89.59 ± 2.99	67.50 ± 7.04	96.70 ± 1.50	93.29 ± 1.44	86.28 ± 6.25	66.01 ± 17.25	**98.03 ± 2.61**	80.45 ± 18.47
19	86.58 ± 7.06	70.07 ± 9.45	64.78 ± 5.03	85.59 ± 2.56	85.30 ± 1.03	67.29 ± 14.74	58.81 ± 17.89	83.61 ± 7.44	**89.01 ± 4.91**
20	88.45 ± 15.68	80.01 ± 4.11	42.51 ± 9.36	84.58 ± 4.77	**96.12 ± 1.52**	84.83 ± 18.81	78.27 ± 16.66	86.85 ± 15.82	82.74 ± 15.58
21	93.36 ± 12.71	84.28 ± 10.84	25.01 ± 4.14	64.82 ± 7.25	88.21 ± 4.01	92.35 ± 6.20	83.90 ± 8.65	**98.19 ± 2.12**	89.47 ± 10.01
22	89.06 ± 1.50	78.17 ± 11.36	58.97 ± 6.53	80.79 ± 3.69	89.11 ± 1.13	89.48 ± 6.02	87.39 ± 7.71	93.40 ± 5.91	**95.09 ± 4.88**
OA(%)	82.26 ± 2.07	72.45 ± 1.06	73.49 ± 1.54	82.60 ± 1.21	79.20 ± 1.23	77.68 ± 4.27	74.62 ± 5.65	83.18 ± 2.28	**88.11 ± 0.96**
AA(%)	80.79 ± 2.43	74.45 ± 1.56	71.93 ± 1.64	79.54 ± 1.93	76.17 ± 0.52	75.67 ± 3.02	68.41 ± 2.02	82.29 ± 1.21	**83.92 ± 1.74**
$k \times 100$	78.20 ± 2.44	66.79 ± 1.52	70.78 ± 1.32	78.55 ± 1.36	74.42 ± 1.37	69.64 ± 8.23	68.80 ± 5.94	79.33 ± 2.60	**83.43 ± 1.92**

On the more challenging WHU-Hi series datasets, including WHU-Hi-HongHu, WHU-Hi-LongKou, and WHU-Hi-HanChuan, which contain fine-grained spectral signatures, high spatial resolution, and complex background interference, the proposed method continues to demonstrate strong competitiveness. For WHU-Hi-HongHu, our OA reaches 88.11%, which clearly exceeds the 83.18% obtained by SSEFN. The AA increases to 83.92%, far above the 68.41% achieved

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TABLE VII

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR WHU-HI-LONGKOU (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	CEGCN	GTFN	CLOLN	S^{2}GFormer	GSC-VIT	MambaHSI	NESSGGCN	SSEFN	Our
1	98.71 ± 1.47	91.20 ± 6.31	85.80 ± 0.66	95.58 ± 1.83	**100.00 ± 0.00**	98.65 ± 1.05	81.62 ± 15.69	98.40 ± 1.43	97.74 ± 2.78
2	83.82 ± 16.17	80.20 ± 8.43	83.91 ± 2.40	84.30 ± 2.11	94.18 ± 0.56	91.66 ± 7.88	87.70 ± 8.92	92.13 ± 4.13	**94.68 ± 4.29**
3	99.67 ± 0.15	95.21 ± 2.50	98.60 ± 1.06	98.21 ± 1.29	99.64 ± 0.13	92.08 ± 4.35	86.40 ± 10.48	99.50 ± 0.46	**99.90 ± 0.17**
4	86.80 ± 2.68	65.24 ± 10.95	78.22 ± 5.71	68.84 ± 4.67	91.29 ± 1.95	75.13 ± 11.12	75.85 ± 11.02	83.52 ± 6.48	**93.69 ± 3.67**
5	**99.58 ± 0.30**	84.33 ± 10.43	97.54 ± 0.45	96.54 ± 1.50	92.64 ± 1.78	68.59 ± 15.69	63.33 ± 20.80	88.09 ± 10.49	91.54 ± 14.42
6	95.26 ± 5.13	87.16 ± 8.00	89.76 ± 3.32	95.35 ± 2.73	90.76 ± 1.98	95.45 ± 2.82	73.61 ± 11.71	**99.24 ± 1.00**	94.30 ± 4.08
7	99.81 ± 0.06	93.73 ± 4.17	81.37 ± 2.43	94.58 ± 0.97	99.20 ± 0.09	99.74 ± 0.35	94.42 ± 3.07	**99.92 ± 0.06**	97.84 ± 2.08
8	78.64 ± 19.00	67.60 ± 11.98	88.34 ± 0.78	**92.15 ± 2.18**	47.06 ± 30.64	70.45 ± 16.25	56.06 ± 17.09	70.39 ± 7.84	88.92 ± 19.09
9	**88.69 ± 3.47**	67.03 ± 13.53	78.80 ± 9.82	77.97 ± 2.87	81.87 ± 10.38	85.64 ± 10.03	55.83 ± 12.64	79.36 ± 4.14	85.28 ± 12.63
OA(%)	93.65 ± 0.90	81.84 ± 4.19	85.29 ± 1.96	88.00 ± 1.44	93.81 ± 1.52	89.25 ± 3.76	81.97 ± 5.12	93.49 ± 2.16	**94.93 ± 1.72**
AA(%)	92.31 ± 1.64	81.30 ± 2.34	86.93 ± 1.28	89.28 ± 1.04	88.52 ± 1.82	86.38 ± 3.52	74.98 ± 3.25	90.06 ± 1.66	**93.77 ± 1.87**
$k \times 100$	91.78 ± 1.15	77.05 ± 2.34	83.06 ± 1.65	85.26 ± 1.76	91.95 ± 1.33	86.50 ± 3.91	76.95 ± 6.31	91.63 ± 2.71	**93.39 ± 2.22**

TABLE VIII

CLASSIFICATION ACCURACIES OBTAINED BY DIFFERENT METHODS FOR WHU-HI-HANCHUAN (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	CEGCN	GTFN	CLOLN	S^{2}GFormer	GSC-VIT	MambaHSI	NESSGGCN	SSEFN	Our
1	86.40 ± 9.95	64.45 ± 1.91	77.78 ± 8.66	86.12 ± 4.33	61.80 ± 6.42	75.13 ± 13.6	67.04 ± 20.89	79.91 ± 8.19	**91.63 ± 3.95**
2	34.56 ± 12.46	49.97 ± 16.81	56.14 ± 11.43	50.09 ± 7.96	54.44 ± 8.42	46.85 ± 9.53	47.96 ± 23.43	**57.67 ± 9.84**	52.32 ± 7.90
3	**86.38 ± 6.03**	60.38 ± 7.27	85.10 ± 1.76	74.01 ± 5.74	63.40 ± 6.05	79.79 ± 10.54	70.58 ± 18.84	75.46 ± 6.81	85.80 ± 6.80
4	97.67 ± 1.18	96.62 ± 7.27	97.74 ± 0.89	97.55 ± 1.46	93.69 ± 2.28	94.26 ± 4.02	87.79 ± 12.98	**97.89 ± 0.52**	97.61 ± 1.62
5	99.81 ± 0.26	80.20 ± 9.27	**99.92 ± 0.07**	87.52 ± 8.31	95.63 ± 3.22	95.25 ± 3.41	80.70 ± 23.74	96.54 ± 3.92	96.12 ± 6.30
6	63.95 ± 14.46	35.42 ± 4.65	58.28 ± 10.89	53.98 ± 4.82	40.59 ± 5.96	41.97 ± 10.33	49.51 ± 15.66	**65.03 ± 6.55**	60.10 ± 5.05
7	89.21 ± 8.11	82.85 ± 8.46	86.42 ± 6.99	**97.09 ± 1.16**	91.16 ± 2.34	84.40 ± 12.94	66.50 ± 25.35	96.24 ± 3.56	90.07 ± 5.86
8	58.43 ± 11.45	33.23 ± 6.23	**65.99 ± 5.15**	44.29 ± 8.65	52.12 ± 1.05	46.26 ± 6.49	49.51 ± 13.20	54.76 ± 5.57	59.93 ± 14.42
9	49.78 ± 17.27	31.70 ± 11.06	45.10 ± 3.83	48.25 ± 4.75	39.19 ± 2.29	48.47 ± 10.50	28.19 ± 9.93	60.30 ± 7.93	**61.68 ± 7.03**
10	87.04 ± 13.12	74.59 ± 17.06	82.54 ± 6.54	**95.06 ± 5.13**	82.79 ± 6.72	74.77 ± 15.38	58.37 ± 9.99	77.71 ± 13.71	83.98 ± 9.48
11	77.22 ± 7.90	70.41 ± 17.63	77.08 ± 18.59	58.44 ± 9.07	**80.58 ± 1.25**	70.39 ± 13.97	58.96 ± 21.05	77.38 ± 16.08	78.83 ± 15.88
12	75.89 ± 12.58	64.16 ± 4.00	72.76 ± 18.89	60.84 ± 10.53	63.27 ± 12.57	72.55 ± 11.78	**79.07 ± 11.40**	67.39 ± 6.65	78.17 ± 8.09
13	52.07 ± 13.33	29.91 ± 1.93	47.33 ± 17.88	36.64 ± 4.25	**62.32 ± 2.57**	55.04 ± 6.21	37.16 ± 11.02	47.00 ± 7.86	55.75 ± 12.88
14	53.39 ± 18.08	52.98 ± 5.68	46.72 ± 16.26	67.11 ± 5.02	62.50 ± 3.97	57.35 ± 9.35	42.89 ± 21.01	**78.96 ± 7.14**	54.51 ± 8.03
15	74.37 ± 9.61	69.14 ± 10.90	78.79 ± 4.92	77.60 ± 2.84	**84.72 ± 4.31**	75.42 ± 9.95	71.26 ± 10.48	79.79 ± 10.62	61.61 ± 13.41
16	87.57 ± 8.77	87.77 ± 10.83	80.66 ± 1.19	95.03 ± 1.66	90.06 ± 0.39	88.89 ± 9.78	90.20 ± 8.62	90.76 ± 7.20	**97.94 ± 1.59**
OA(%)	74.46 ± 4.03	65.89 ± 2.53	72.13 ± 3.27	76.41 ± 1.89	71.38 ± 0.17	71.71 ± 4.06	66.31 ± 5.39	76.04 ± 3.59	**80.59 ± 1.58**
AA(%)	73.36 ± 3.17	61.49 ± 0.66	72.40 ± 3.91	70.60 ± 1.27	69.89 ± 1.22	69.17 ± 1.64	61.61 ± 3.97	74.02 ± 1.82	**75.38 ± 1.57**
$k \times 100$	70.66 ± 4.49	60.89 ± 2.57	68.11 ± 3.73	72.58 ± 2.16	67.18 ± 0.24	67.42 ± 4.48	60.97 ± 5.86	72.40 ± 3.99	**77.27 ± 1.88**

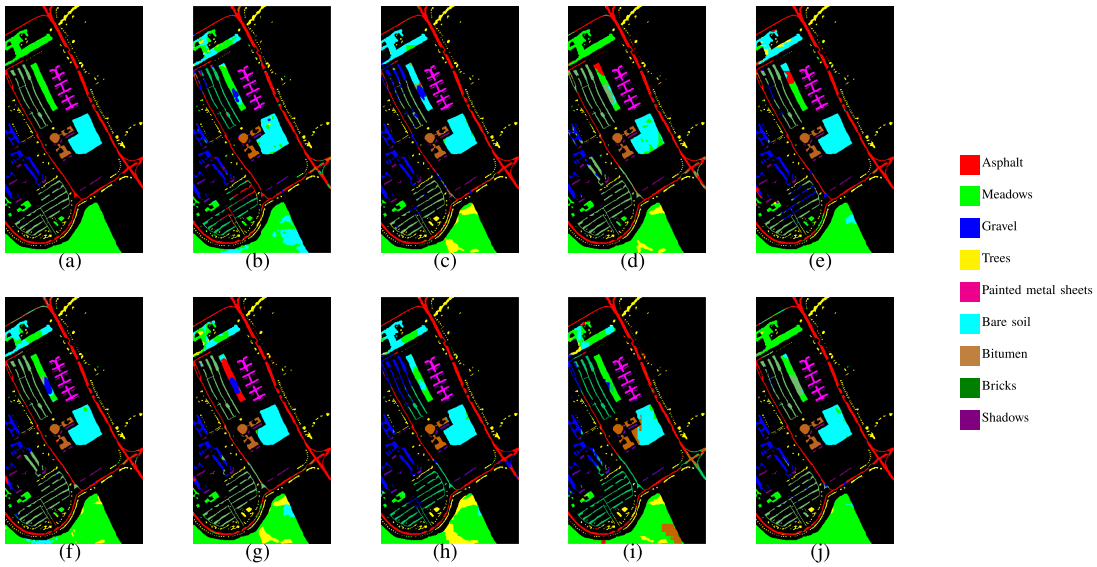


Fig. 9. Qualitative visualization of the classification map for the PaviaU dataset. (a) Ground-truth map. (b) CEGCN. (c) GTFN. (d) CLOLN. (e) S^{2}GFormer. (f) GSC-VIT. (g) MambaHSI. (h) NESSGGCN. (i) SSEFN. (j) Our.

by NESSGGCN. The Kappa coefficient rises to 83.43%, reflecting significantly stronger consistency between the predicted labels and the ground truth. For WHU-Hi-LongKou, our method achieves an OA of 94.93%, an AA of 93.77%, and a Kappa coefficient of 93.39%. The accuracies for Class 2 and Class 3 reach 94.68% and 99.90%, indicating stable performance across different land-cover categories. On WHU-Hi-HanChuan, our method also achieves the best overall

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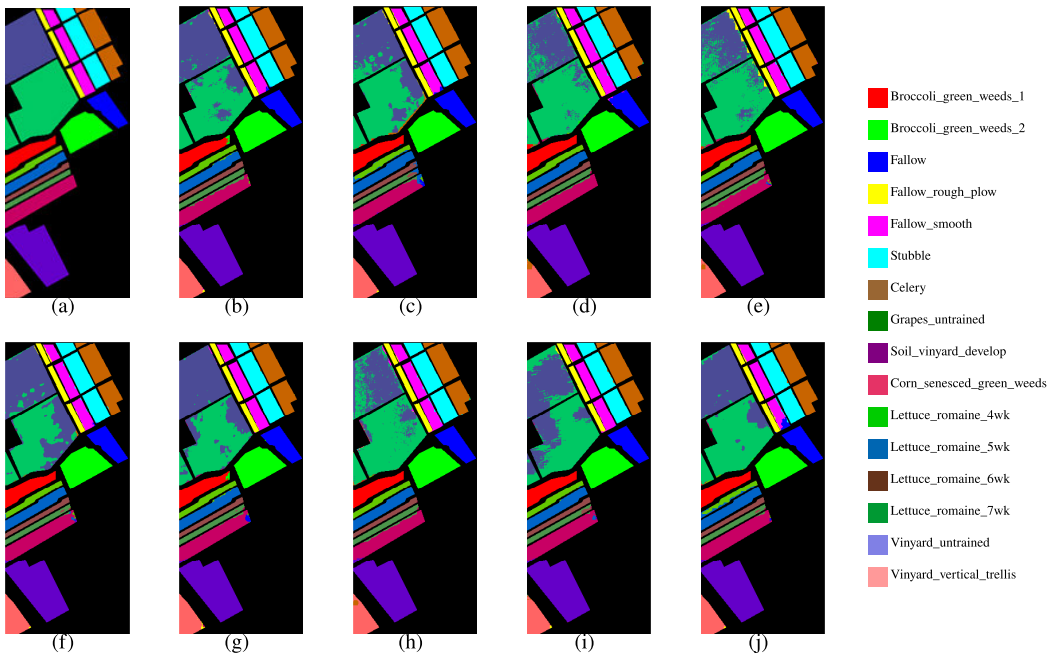


Fig. 10. Qualitative visualization of the classification map for the Salinas dataset. (a) Ground-truth map. (b) CEGCN. (c) GTFN. (d) CLOLN. (e) S^2GFormer. (f) GSC-VIT. (g) MambaHSI. (h) NESSGGCN. (i) SSEFN. (j) Our.

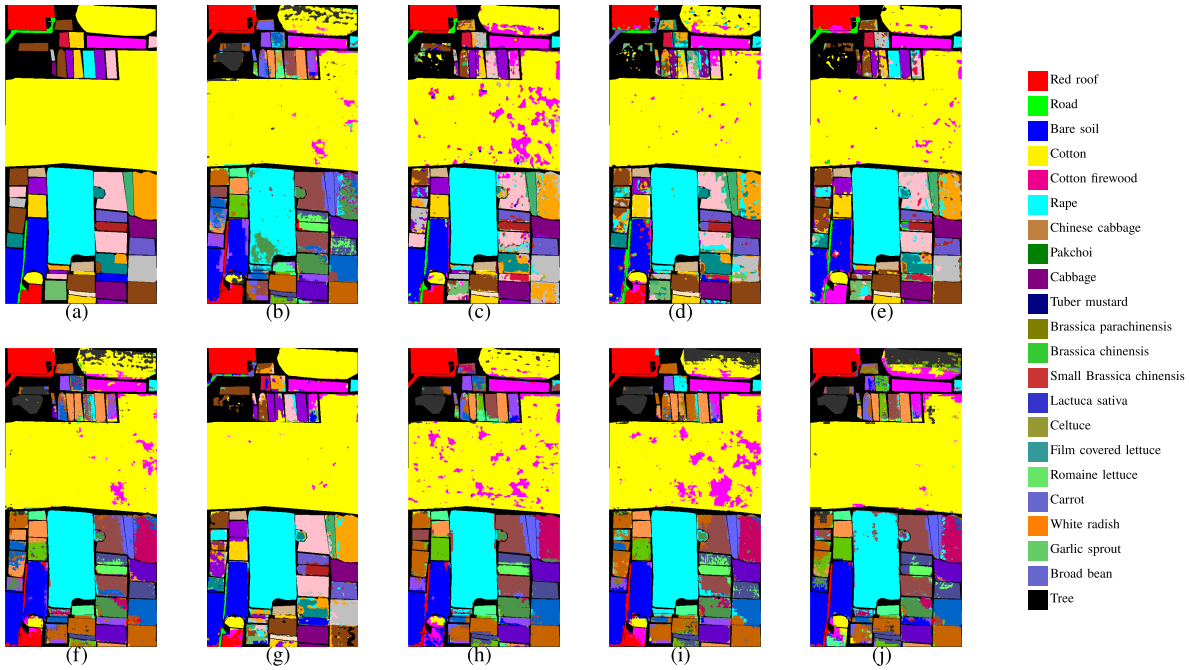


Fig. 11. Qualitative visualization of the classification map for the WHU-Hi-HongHu dataset. (a) Ground-truth map. (b) CEGCN. (c) GTFN. (d) CLOLN. (e) S^2GFormer. (f) GSC-VIT. (g) MambaHSI. (h) NESSGGCN. (i) SSEFN. (j) Our.

performance, with an OA of 80.59%, an AA of 75.38%, and a Kappa coefficient of 77.27%. Especially in category 1, its accuracy rate reaches 91.63%, and it also achieves good results in several other categories that are difficult to classify. Although some other methods produce visually comparable homogeneous regions, their classification maps still contain more isolated errors and less accurate boundary delineation. As shown in Figs. 11–13, our method generates smoother classification results with fewer isolated misclassifications and clearer object boundaries across these challenging UAV hyperspectral scenes.

In summary, the comparative experiments conducted on five representative benchmark datasets provide strong evidence of the effectiveness of the proposed method. The model exhibits outstanding stability, excellent generalization ability, and superior visual coherence across diverse and challenging classification scenarios. Overall, the results confirm that the proposed framework surpasses existing approaches across

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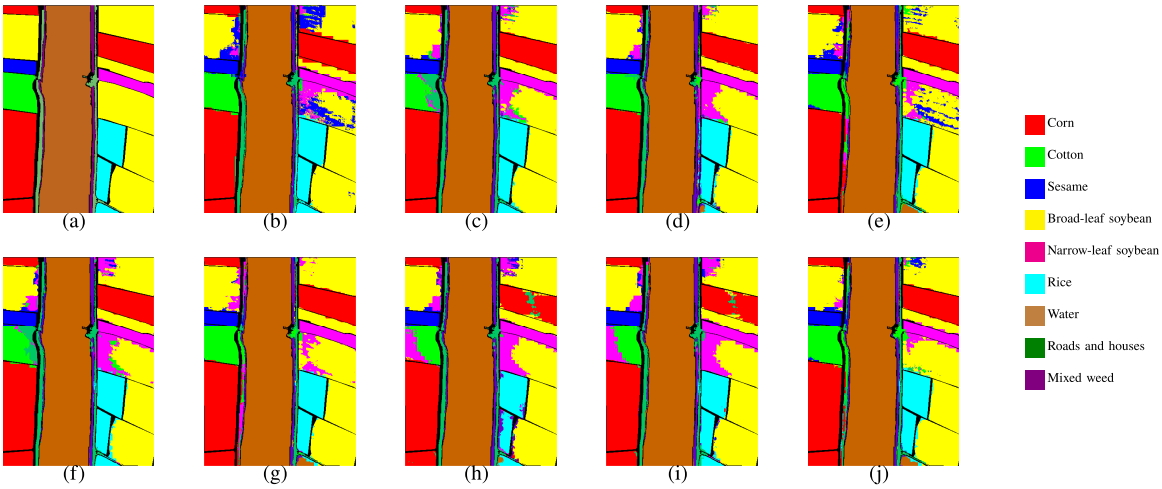


Fig. 12. Qualitative visualization of the classification map for the WHU-Hi-LongKou dataset. (a) Ground-truth map. (b) CEGCN. (c) GTFN. (d) CLOLN. (e) S^2GFormer. (f) GSC-VIT. (g) MambaHSI. (h) NESSGGCN. (i) SSEFN. (j) Our.

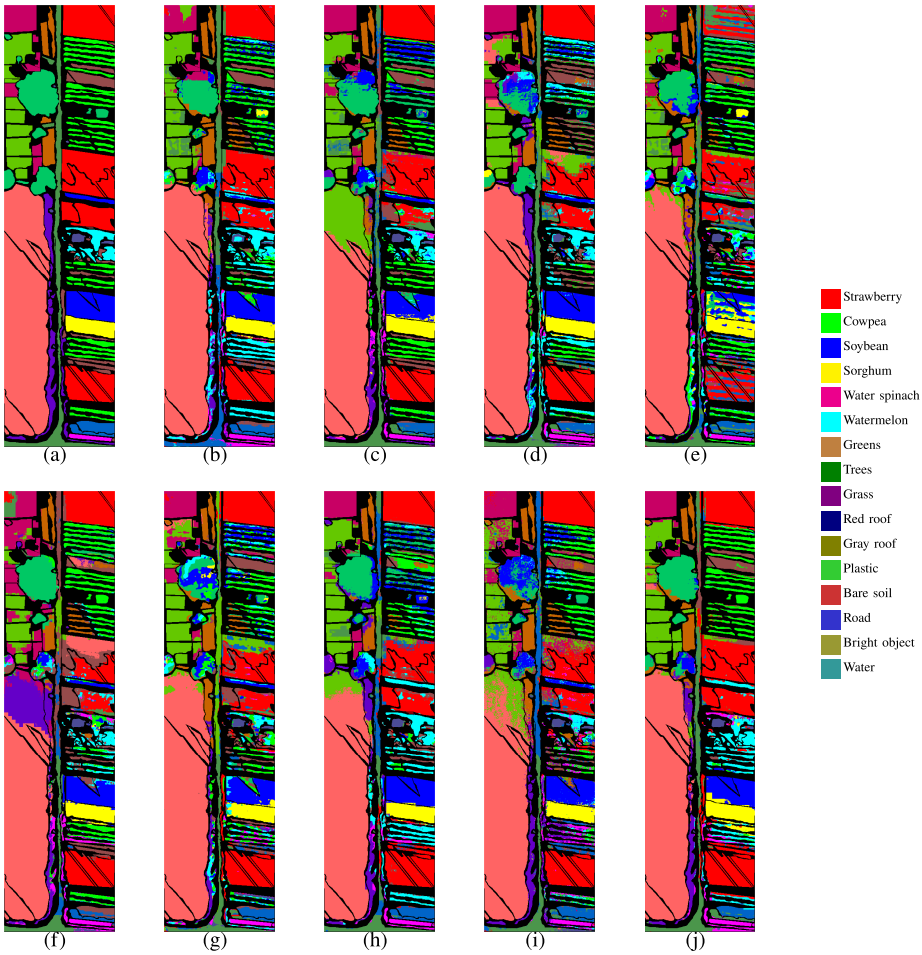


Fig. 13. Qualitative visualization of the classification map for the WHU-Hi-HanChuan dataset. (a) Ground-truth map. (b) CEGCN. (c) GTFN. (d) CLOLN. (e) S^2GFormer. (f) GSC-VIT. (g) MambaHSI. (h) NESSGGCN. (i) SSEFN. (j) Our.

multiple evaluation metrics and offers a more reliable and practical solution for hyperspectral image classification.

IV. Conclusion

In this article, we propose a method that aims to address several existing challenges in the field of hyperspectral image classification. By integrating superpixel-level global structural information and pixel-level local detail features, the proposed approach effectively improves the accuracy and robustness of few-shot hyperspectral image classification tasks. The MFAM is capable of fusing three types of information: structural topology, spatial location, and feature similarity. It extracts multiscale graph structural features using differentiated Top-k operations and subsequently generates an adjacency

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matrix with stronger global representation and detailed description capabilities, accurately capturing the complex relationships among nodes. The synergistic integration of the GPM and the CAM not only enhances the aggregation of node features but also encourages the model to focus on key discriminative features. Meanwhile, the global supervision strategy enables efficient fusion of local detail information and global semantic information. In the upper branch, we utilize depthwise separable convolutions combined with attention mechanisms to extract pixel-level spatial features through multiscale convolution kernels. Furthermore, the incorporation of channel attention and position attention mechanisms strengthens the model's ability to perceive local textures and structural details. Experimental results demonstrate that, compared with other state-of-the-art methods, IPGCNet significantly improves the OA on five datasets. The average per-class accuracy also shows stable performance across different datasets and varying numbers of labeled samples.

Although the proposed framework performs well on multiple datasets, inaccurate segmentation may have an impact on the construction of the adjacency matrix and the modeling of relationships. Moreover, the current evaluation mainly focuses on a small number of sample settings within the dataset, and there is insufficient exploration of generalization capabilities across scenarios and modalities. Future research will explore integrating pretrained models into our proposed method by leveraging large-scale pretraining to provide richer prior knowledge, thereby enhancing the generalization ability across scenarios and modalities, while maintaining the structural advantages of graph-based reasoning.

REFERENCES

[1] L. Duponchel, "Exploring hyperspectral imaging data sets with topological data analysis," *Analytica Chim. Acta*, vol. 1000, pp. 123–131, Feb. 2018. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0003267017313077

[2] N. Audebert, B. Le Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag. (replaces Newsletter)*, vol. 7, no. 2, pp. 159–173, Jun. 2019.

[3] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag. (replaces Newsletter)*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[4] Y. Jiang, Y. Li, S. Zou, H. Zhang, and Y. Bai, "Hyperspectral image classification with spatial consistence using fully convolutional spatial propagation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10425–10437, Dec. 2021.

[5] S. Wang, Y. Bi, J. Du, T. Zhang, X. Gao, and E. Jin, "The unmanned aerial vehicle (UAV)-based hyperspectral classification of desert grassland plants in inner mongolia, China," *Appl. Sci.*, vol. 13, no. 22, p. 12245, Nov. 2023. [Online]. Available: https://www.mdpi.com/2076-3417/13/22/12245

[6] B. Zhang, Y. Chen, Z. Li, S. Xiong, and X. Lu, "SANet: A self-attention network for agricultural hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–15, 2024.

[7] X. Wang, J. Liu, W. Chi, W. Wang, and Y. Ni, "Advances in hyperspectral image classification methods with small samples: A review," *Remote Sens.*, vol. 15, no. 15, p. 3795, Jul. 2023. [Online]. Available: https://www.mdpi.com/2072-4292/15/15/3795

[8] W. Chen, S. Ouyang, J. Yang, X. Li, G. Zhou, and L. Wang, "JAGAN: A framework for complex land cover classification using gaofen-5 AHSI images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1591–1603, 2022.

[9] S. Tang et al., "Synergistic spectral–spatial fusion in hyperspectral imaging: Dual attention-based Rice seed varieties identification," *Food Control*, vol. 176, Oct. 2025, Art. no. 111411. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0956713525002804

[10] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

[11] C. Fu, B. Du, and L. Zhang, "ReSC-Net: Hyperspectral image classification based on attention-enhanced residual module and spatial-channel attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5518615.

[12] X. Tian, F. Luo, C. Fu, T. Guo, B. Du, and J. Chanussot, "MAGS: Max-gap loss-guided siamese-reconstruction network for hyperspectral image partial label learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5506813.

[13] C. Fu, T. Zhou, T. Guo, Q. Zhu, F. Luo, and B. Du, "CNN-transformer and channel-spatial attention based network for hyperspectral image classification with few samples," *Neural Netw.*, vol. 186, Jun. 2025, Art. no. 107283.

[14] C. Chen et al., "Hyperspectral classification based on spectral–spatial convolutional neural networks," *Eng. Appl. Artif. Intell.*, vol. 68, pp. 165–171, Feb. 2018.

[15] J. Xia, L. Bombrun, T. Adali, Y. Berthoumieu, and C. Germain, "Spectral–spatial classification of hyperspectral images using ICA and edge-preserving filter via an ensemble strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4971–4982, Aug. 2016.

[16] K. Boukhechba, H. Wu, and R. Bazine, "DCT-based preprocessing approach for ICA in hyperspectral data analysis," *Sensors*, vol. 18, no. 4, p. 1138, Apr. 2018. [Online]. Available: https://www.mdpi.com/1424-8220/18/4/1138

[17] T. Lu, M. Liu, W. Fu, and X. Kang, "Grouped multi-attention network for hyperspectral image spectral–spatial classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507912.

[18] J. Zhou, S. Zeng, G. Gao, Y. Chen, and Y. Tang, "A novel spatial–spectral pyramid network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023, Art. no. 5519314.

[19] D. Liao, C. Shi, and L. Wang, "A spectral–spatial fusion transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515216.

[20] B. Liu, X. Yu, A. Yu, P. Zhang, and G. Wan, "Spectral–spatial classification of hyperspectral imagery based on recurrent neural networks," *Remote Sens. Lett.*, vol. 9, no. 12, pp. 1118–1127, Dec. 2018, doi: 10.1080/2150704x.2018.1511933.

[21] W. Zhou, S.-i. Kamata, H. Wang, and X. Xue, "Multiscanning-based RNN-transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512319.

[22] Z. Li, W. Huang, L. Wang, Z. Xin, and Q. Meng, "CNN and transformer interaction network for hyperspectral image classification," *Int. J. Remote Sens.*, vol. 44, no. 18, pp. 5548–5573, Sep. 2023, doi: 10.1080/01431161.2023.2249598.

[23] Z. Zhao, X. Xu, J. Li, S. Li, and A. Plaza, "Gabor-modulated grouped separable convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518817.

[24] J. Li, X. Zhao, Y. Li, Q. Du, B. Xi, and J. Hu, "Classification of hyperspectral imagery using a new fully convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 292–296, Feb. 2018.

[25] H. Gong et al., "Multiscale information fusion for hyperspectral image classification based on hybrid 2D-3D CNN," *Remote Sens.*, vol. 13, no. 12, p. 2268, 2021. [Online]. Available: https://www.mdpi.com/2072-4292/13/12/2268

[26] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[27] Y. Xu, D. Wang, L. Zhang, and L. Zhang, "Dual selective fusion transformer network for hyperspectral image classification," *Neural Netw.*, vol. 187, Jul. 2025, Art. no. 107311. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S089360802500190X

[28] F. Luo, Y. Hua, C. Fu, T. Guo, Y. Duan, and G. Shi, "MMD-MLP: LiDAR-guided hyperspectral data classification using local–global directional-MLP with multi-resolution multi-scale representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–14, 2025, Art. no. 5508414.

[29] Y. Xu, D. Wang, H. Jiao, L. Zhang, and L. Zhang, "MambaMoE: Mixture-of-spectral–spatial-experts state space model for hyperspectral image classification," *Inf. Fusion*, vol. 127, Mar. 2026, Art. no. 103811.

[30] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[31] M. Mendieta, B. Han, X. Shi, Y. Zhu, and C. Chen, "Towards geospatial foundation models via continual pretraining," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16760–16770.

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[32] S. Fu, W. Liu, K. Zhang, and Y. Zhou, "Example-feature graph convolutional networks for semi-supervised classification," *Neurocomputing*, vol. 461, pp. 63–76, Oct. 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0925231221010924

[33] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[34] S. Pu, Y. Wu, X. Sun, and X. Sun, "Hyperspectral image classification with localized graph convolutional filtering," *Remote Sens.*, vol. 13, no. 3, p. 526, 2021. [Online]. Available: https://www.mdpi.com/2072-4292/13/3/526

[35] Y. Ding, Y. Guo, Y. Chong, S. Pan, and J. Feng, "Global consistent graph convolutional network for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 70, 2021, Art. no. 5501516.

[36] S. Jia, S. Jiang, S. Zhang, M. Xu, and X. Jia, "Graph-in-graph convolutional network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1157–1171, Jan. 2024.

[37] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.

[38] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508614.

[39] G. Dai, X. Wang, X. Zou, C. Liu, and S. Cen, "Mrgat: Multi-relational graph attention network for knowledge graph completion," *Neural Netw.*, vol. 154, pp. 234–245, Oct. 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0893608022002714

[40] B. Xi et al., "Semisupervised cross-scale graph prototypical network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9337–9351, Nov. 2023.

[41] K. Xu, Y. Zhao, L. Zhang, C. Gao, and H. Huang, "Spectral–spatial residual graph attention network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 5509305.

[42] H. Zheng, H. Su, Z. Wu, M. E. Paoletti, and Q. Du, "Graph convolutional network with relaxed collaborative representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5533513.

[43] G. Xu, S. Yang, and Z. Feng, "Dual-semantic graph convolution network for hyperspectral image classification with few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5508815.

[44] A. Magner, M. Baranwal, and A. O. Hero, "Fundamental limits of deep graph convolutional networks for graph classification," *IEEE Trans. Inf. Theory*, vol. 68, no. 5, pp. 3218–3233, May 2022.

[45] W. Wang, C. Liu, G. Liu, and X. Wang, "Cf-gcn: Graph convolutional network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5607013.

[46] Y. Han, K. Park, Y. Jung, and L.-S. Kim, "EGCN: An efficient GCN accelerator for minimizing off-chip memory access," *IEEE Trans. Comput.*, vol. 71, no. 12, pp. 3127–3139, Dec. 2022.

[47] Q. Diao, Y. Dai, C. Zhang, Y. Wu, X. Feng, and F. Pan, "Superpixel-based attention graph neural network for semantic segmentation in aerial images," *Remote Sens.*, vol. 14, no. 2, p. 305, 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/2/305

[48] H. Cy, "Improved SLIC image segmentation algorithm based on k-means," *Cluster Comput.*, vol. 20, pp. 1017–1023, Feb. 2017.

[49] B. Kunkel, F. Blechinger, D. Viehmann, H. van der Piepen, and R. Doerffer, "ROSIS imaging spectrometer and its potential for ocean parameter measurements (airborne and space-borne)," *Int. J. Remote Sens.*, vol. 12, no. 4, pp. 753–761, Apr. 1991, doi: 10.1080/01431169108929691.

[50] G. Vane, R. O. Green, T. G. Chrien, H. T. Enmark, E. G. Hansen, and W. M. Porter, "The airborne visible/infrared imaging spectrometer (aviris)," *Remote Sens. Environ.*, vol. 44, no. 2, pp. 127–143, 1993. [Online]. Available: https://www.sciencedirect.com/science/article/pii/003442579390012M

[51] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.

[52] A. Yang, M. Li, Y. Ding, D. Hong, Y. Lv, and Y. He, "GTFN: GCN and transformer fusion network with spatial–spectral features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 6600115.

[53] C. Li, B. Rasti, X. Tang, P. Duan, J. Li, and Y. Peng, "Channel-layer-oriented lightweight spectral–spatial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504214.

[54] S. Huang et al., "S^2GFormer: A transformer and graph convolution combining framework for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537514.

[55] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.

[56] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "Mambahsi: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[57] Z. Zhang, L. Huang, B.-H. Tang, Q. Wang, Z. Ge, and L. Jiang, "Non-Euclidean spectral–spatial feature mining network with gated GCN-CNN for hyperspectral image classification," *Expert Syst. Appl.*, vol. 272, May 2025, Art. no. 126811. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417425004336

[58] S. Liu, C. Fu, Y. Duan, X. Wang, and F. Luo, "Spatial–spectral enhancement and fusion network for hyperspectral image classification with few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5502414.

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